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WRFER WORLD RESEARCH FORUM
FOR ENGINEERS AND RESEARCHERS

WRFER INTERNATIONAL CONFERENCE

28th June 2025

In Association with



WRFASE
World Research Forum for Advances
in Science and Engineering



SARC
South Asian Research Center

PUNE, INDIA

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Organized by

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EDITORIAL

It is my proud privilege to welcome you all to the International Conference at Pune, India. I am happy to see the papers from all part of the world and some of the best paper published in this proceedings. This proceeding brings out the various Research papers from diverse areas of Science, Engineering, Technology and Management. This platform is intended to provide a platform for researchers, educators and professionals to present their discoveries and innovative practice and to explore future trends and applications in the field Science and Engineering. However, this conference will also provide a forum for dissemination of knowledge on both theoretical and applied research on the above said area with an ultimate aim to bridge the gap between these coherent disciplines of knowledge. Thus the forum accelerates the trend of development of technology for next generation. Our goal is to make the Conference proceedings useful and interesting to audiences involved in research in these areas, as well as to those involved in design, implementation and operation, to achieve the goal.

I once again give thanks to the IRAJ International, WRFASE & SARC for organizing this event in Pune, India. I am sure the contributions by the authors shall add value to the research community. I also thank all the International Advisory members and Reviewers for making this event a Successful one.

Editor-In-Chief

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AI-DRIVEN PLACEMENT PREDICTION AND INSIGHTS

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Abstract - A comprehensive, data-driven platform called "AI-Driven Placement Prediction and Insights" was created to help academic institutions and students better understand and improve placement preparedness. The system combines a number of contemporary technologies to provide precise placement forecasts and useful information. Flutter Web was used in the development of the user interface, making it responsive and fluid.

Primarily, the platform utilizes a Random Forest algorithm-based machine learning model that was trained on a variety of student characteristics, such as academic performance (10th, 12th, and graduation grades), quiz scores, technical scores (out of 20), logical reasoning, verbal ability, quantitative aptitude, the number of projects and internships completed, and mastery of critical skills like SQL, DSA, Java, Python, C++, ML, AI, GenAI, Tableau, ReactJS, NodeJS, and more. Real-time predictions based on the most recent student inputs are ensured by exposing the model through a Flask API that is implemented on Render.

The creation of comprehensive Power BI dashboards is made possible by the system's safe management and storage of all data. These dashboards help administrators and students identify placement trends, skill gaps, and overall performance data by providing relevant visuals.

To ensure scalability and accessibility, the complete system is integrated into the Flutter Web application, and Firebase Hosting is used for the final release. This project demonstrates the usefulness of AI and ML in the educational field as well as the ability of full-stack integration to provide insightful, engaging, and significant solutions.

Keywords - Placement Analysis, Placement Prediction, Machine Learning, Data Visualization, Data-Driven

I. INTRODUCTION

In today's competitive job market, student placements play a critical role in shaping the future of graduates and determining the reputation of educational institutions. However, predicting placement outcomes and identifying the factors influencing employability remain challenging tasks due to the dynamic nature of industry demands and student profiles.

This project, Placement Analysis and Prediction Using Machine Learning, addresses these challenges by leveraging the power of data science and predictive modeling.

By analyzing historical placement data and current student attributes, the system provides actionable insights into placement trends and predicts the likelihood of students securing job offers. The project aims to empower students and academic administrators to make informed decisions by detecting talent shortages and aligning their efforts with industry requirements.

The system incorporates machine learning algorithms to analyze features such as programming skills, academic performance, extracurricular involvement, and recruitment patterns. Through visualizations and predictive analytics, the platform offers a user-friendly interface that highlights trends and provides

recommendations for improvement. Additionally, the tool ensures scalability, data security, and ease of access, making it suitable for diverse educational institutions.

This project bridges the gap between academic preparation and industry expectations, enabling institutions to enhance their placement strategies and helping students unlock their career potential.

II. HELPFUL LITERATURE REVIEW

The development of predictive systems for student placements has seen significant advancements in recent years, primarily due to the application of machine learning (ML) techniques. Researchers like Milind Ruparel et al. (2024) and Advait Joshi et al. (2024) have demonstrated the effectiveness of supervised learning algorithms such as Logistic Regression, Random Forest, and Decision Trees in predicting placement probabilities. These algorithms leverage features like academic performance, technical skills, and extracurricular activities to enhance prediction accuracy.

Several studies emphasize the importance of feature selection and data preprocessing in improving model performance. For instance, Rushikesh Joshi et al. (2024) highlighted how including factors like internships and project work can significantly refine predictions. Similarly, Vinod Agrawal et al. (2023)

reviewed various ML algorithms, finding Decision Trees and Random Forests particularly effective in placement prediction.

Moreover, modern predictive systems aim to go beyond mere prediction by offering actionable insights. Priyanka Shahane (2022) and Varsha Harihar (2020) underscored the importance of integrating visualization tools and analytical features to identify skill gaps and recommend interventions. These systems are not limited to predictions but also provide strategic guidance to institutions for enhancing placement outcomes.

The literature collectively underscores the potential of ML-driven predictive tools in transforming traditional placement processes into data-driven frameworks. This lays a strong foundation for the proposed "Placement Analysis and Prediction" project, which aims to implement a robust and accurate system tailored for real-world educational applications.

III. METHODOLOGY

The implementation of the Placement Prediction and Analysis project can be divided into several key phases, each encompassing specific tasks and objectives. These phases guide the systematic development and deployment of the project:

1. Data Collection

- Sources: Historical placement data from the college's database, including:
- Academic scores (e.g., GPA).
- Technical skills and certifications.
- Internship and project details.
- Extracurricular activities.
- Placement outcomes.
- Format: Data is collected in a structured format (e.g., CSV files).

2. Data Preprocessing

- **Data Cleaning:**
- Handling missing values (e.g., using mean/mode imputation).
- Removing or correcting outliers.

- **Data Transformation:**

Encoding categorical data (e.g., programming languages known) into numerical form.

- **Feature Engineering:**

Recognizing and developing novel traits that might affect placement results.

3. Model Development

- **Algorithm Selection:**

The selection of algorithms for classification problems is dependent on factors including Neural

Networks, Random Forests, Decision Trees, and Logistic Regression.

- **Model Training:**

- Splitting the data into training (80%) and testing (20%) sets.
- Using cross-validation to ensure robustness and avoid overfitting.

4. Model Evaluation

- **Metrics:** Evaluating the models using metrics like:
- Accuracy
- Precision
- Recall
- F1-score
- **Comparison:** Comparing the performance of various models to select the best-performing one for deployment.

5. Insights and Recommendations:

Branch performance insights

- Evaluates effectiveness of individual branches.
- Identifies top performers and areas for improvement.

Yearly placement analysis

- Examines placement trends and statistics across different years.
- Reveals long-term patterns and growth in placements.

Company insights: leading placement volume

- Identifies companies recruiting the most students (placement volume).

Uncovering trends by location, department, And role:

- Analyzes placement patterns based on geographic location, academic department, and specific job roles.
- Helps understand skill demand and curriculum needs.

6. Visualization and Reporting

Visualization Tools: Creating dashboards to display placement probabilities, influencing factors, and trends using tools like PowerBI or Python visualization libraries.

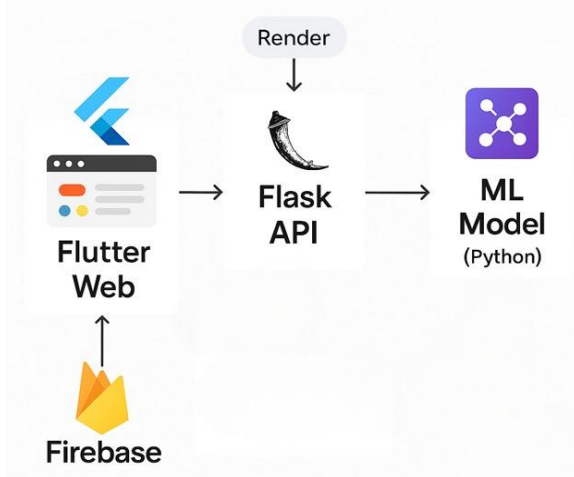
Reports: Generating reports summarizing placement insights and predictions.

7. Deployment

- Platform: Deploying the system as a web or mobile application for ease of access.
- Tools: Using platforms like Flask, Flutter for web deployment, with cloud support (Render).

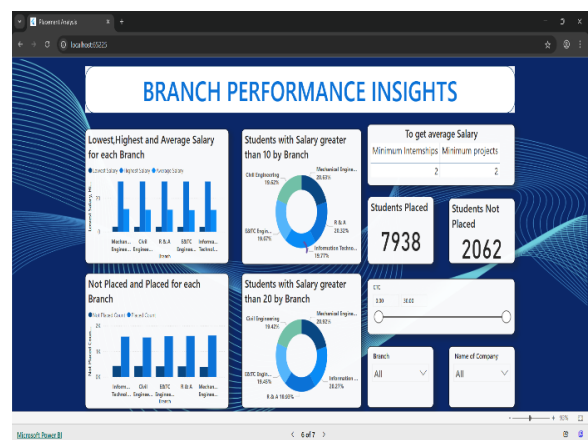
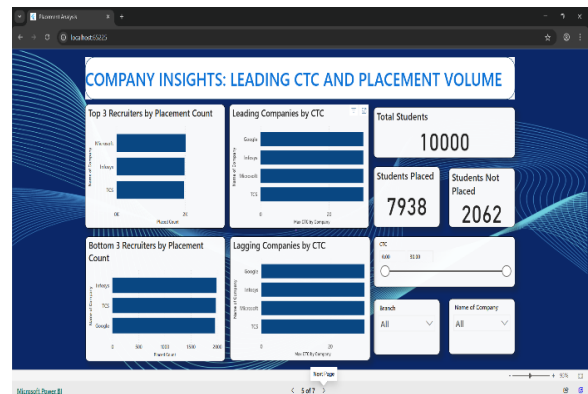
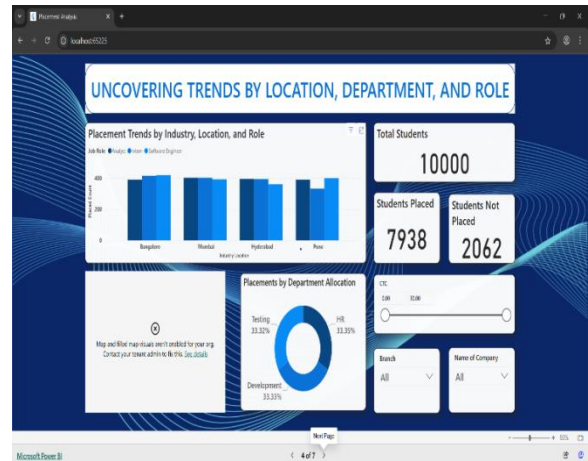
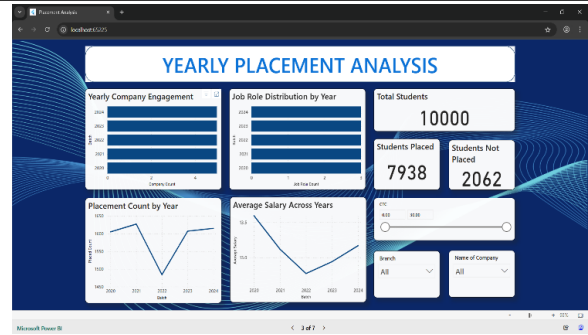
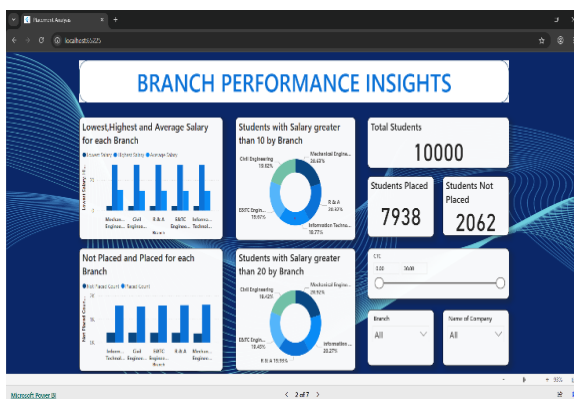
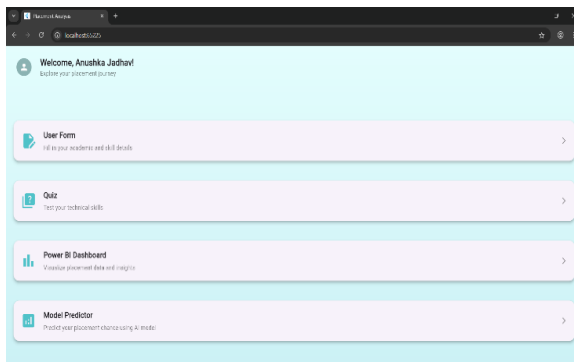
8. Testing and Validation

- User Testing: Collecting feedback from students and administrators to refine the system.
- Performance Validation: Ensuring the model performs well on unseen data and aligns with real-world requirements.



IV. RESULTS

Random Forest achieved a perfect accuracy of 100%, with precision, recall, and F1-score all equal to 1.00 for both classes. While this may indicate strong predictive capability, such perfect results could also suggest possible overfitting, especially in real-world applications.



User Form

Full Name: Anushka Jadhav
 Email: anushka@gmail.com
 Phone Number: 1234567890
 10th Marks (%): 85
 12th Marks (%): 95
 Number of Internships: 3
 Number of Projects: 5
 Languages (English, Hindi, etc.): English, Hindi, Marathi
 Technical Skills: Python, Java, WebDev, ReactJS
 COPE (0-10): 8

Placement Prediction

Test Results

Tech Metrics

Metric	Score (0-10)
Overall Score	8.5 (85%)
Technical Skills	9.0 (90%)
Soft Skills	8.0 (80%)

Placement Prediction Probability: 58.00%

Recommendations:

- Consider your strengths and weaknesses to identify areas for improvement.
- Enhance your technical skills by learning new technologies.
- Gain practical experience through internships and projects.
- Develop soft skills like communication and teamwork.
- Stay updated with industry trends and technologies.
- Consider further education or certifications in your field.
- Network with professionals in your industry.
- Keep your resume and portfolio up-to-date.

V. CONCLUSION AND FUTUREWORK

The dataset's class imbalance issue is demonstrated by the consistently declining macro F1-scores for all model but Random Forest. Overall, the study shows that Random Forest is the most robust and complete model for this prediction task. The Random Forest classifier outperformed the other machine learning techniques with perfect results in accuracy, precision, recall, and F1-score. This demonstrates its high predictive capacity, but it also raises concerns about potential overfitting, which requires further investigation. The relatively good performance of other models, such as Logistic Regression, XGBoost, and SVM, but their difficulties with class imbalance brought attention to the importance of dataset balancing and model tuning.

The study also highlights how important it is to assess and display model results using intuitive dashboards and analytics. In addition to helping administrators make data-driven strategic choices, these visual aids assist students in identifying their areas of strength and progress. Project experience, internships, programming skills, and academic accomplishment were shown to be significant indications that closely matched the demands of the real industry.

All things considered, this study demonstrates how educational institutions may apply these findings to improve placement outcomes and validates that machine learning is a valid technique for predicting placement prevalence. Because prediction, analysis, and recommendation are all integrated into a single,

coherent platform, this method is both practical and expandable. By closing the gap between academic success and industry readiness, it creates the foundation for more astute, data-driven institutional planning and career advising.

FUTURE WORK

While the current model shows promising results, several avenues exist for further improvement:

1. **Addressing Class Imbalance:** Future iterations should implement advanced techniques such as SMOTE, ADASYN, or cost-sensitive learning to mitigate class imbalance and improve performance on minority classes.
2. **Incorporating Real-Time Data:** Integrating dynamic or real-time student performance data (e.g., ongoing project performance, recent certifications) could enhance prediction accuracy.
3. **Model Interpretability:** Introducing explainable AI (XAI) methods can help users understand the rationale behind predictions, increasing transparency and trust in the system.
4. **Deployment at Scale:** Future development can focus on making the system scalable across multiple institutions, integrating APIs, cloud-based storage, and robust security mechanisms.
5. **Personalized Recommendations:** Enhancing the system to provide personalized roadmaps for individual students based on their unique strengths and weaknesses can significantly boost their placement chances.

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FASHION ADVISOR MODEL BASED ON SKIN TONE AND BODY TYPE

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Abstract - In an era of personalized consumer experiences, the need for tailored fashion advice has grown significantly. This study offers a Fashion Advisor Model created especially for individual customers at a designer studio, offering suggestions based on a mix of body type, skin tone, and cultural fashion tastes. Using machine learning techniques like GANs, CNNs, and decision trees, together with skin color recognition via the HSV color model, the model provides real-time individualized clothing recommendations. It customizes its suggestions based on user profiles using picture recognition and personal input, offering outfit options for both corporate and informal settings. These suggestions are further improved over time by a feedback process, which further increases user happiness and accuracy. This model is a prime example of how sophisticated AI can enhance fashion advice and provide consumers a customized, interactive, and one-of-a-kind style experience.

Keywords - Fashion Recommendation, Skin Tone Detection, Body Type Classification, GANs, Personalized Styling, CNN, HSV Color Mode

I. INTRODUCTION

Personalized fashion recommendations are revolutionizing how customers approach their style, especially in designer studios, where individuality is paramount. Designed to be utilized by individual clients in such circumstances, the proposed Fashion Advisor Model aims to provide a smart styling assistant that respects both the functional and aesthetic elements of fashion. The model considers users' unique skin tones, body types, and cultural preferences in an effort to offer recommendations that not only match with cultural context but also highlight the wearer's natural characteristics. The model classifies skin tone using the Hue, Saturation, Value (HSV) color model, which allows for a sophisticated way to discover undertones that might influence color harmony in clothing choices.

The classification of body types follows common archetypes, such as Rectangle, Inverted Triangle, Hourglass, Apple, Pear, Spoon, and Diamond, to ensure that the costumes produced fit a variety of shapes and sizes. Business dress, casual styles, and culturally acceptable clothing are all discussed to ensure that the recommendations are suitable for a range of situations. Using a variety of methods, including Generative Adversarial Networks (GANs), the Fashion Advisor Model technically visualizes outfits and creates style combinations that satisfy client specifications. Content-based and collaborative filtering techniques are commonly used by recommendation systems to analyze user preferences and enhance recommendations. To increase the accuracy of image recognition for users' skin tone and body type, Convolutional Neural Networks (CNNs) are integrated into the model. Algorithms such as k-Nearest Neighbors (KNN), decision trees,

and random forests, which categorize clothing according to user similarities, further enable personalized recommendations. A mix of human input and image recognition is used to gather user data, providing the model with current and accurate information about each customer's profile. When the model is finished processing, it displays personalized clothing suggestions in the form of created pictures. With an easy-to-use interface powered by HTML, CSS, and JavaScript, and Flask for backend processing, the system offers a seamless user experience.

A feedback mechanism also enables users to respond to and edit their suggestions, promoting ongoing enhancements to the model's precision and applicability. This Fashion Advisor Model demonstrates the possibility of AI-driven fashion guidance in high-end, personalized retail environments. By combining technological proficiency with fashion understanding, it provides customers with personalized, culturally appropriate, and contextually appropriate style solutions.

II. RELATED WORK

Personalized fashion recommendation systems have garnered a lot of interest as current research focuses on enhancing user experience through tailored recommendations based on individual preferences, skin tone, and body form. Previous studies have demonstrated that a variety of methods, including content-based filtering, collaborative filtering, and deep learning approaches, may accomplish customization. Despite these advancements, challenges such as the cold-start problem, data imbalance, and accurately determining body shapes and skin tones persist

1. COLLABORATIVE AND CONTENT-BASED FILTERING

Collaborative filtering and content-based filtering are commonly employed in fashion recommendation systems. Kun- cham et al. (2024) developed a comprehensive fashion recommender system that combines these methods, addressing issues of user satisfaction and preference prediction. Vinitha et al. (2024) incorporated collaborative filtering and real-time behavior analysis in a system that adapts to user preferences. However, both systems face difficulties with new users and require extensive datasets to achieve accuracy.

2. SKIN TONE DETECTION AND COLOR ANALYSIS

Skin tone detection is integral to creating inclusive fashion recommendations, as it allows for color palettes that enhance an individual's natural complexion. Studies by Chakraborty et al. (2021) and Daithankar et al. (2014) examined various color spaces, such as RGB and HSV, to optimize skin tone detection in recommendation models. Their findings underscore the efficacy of CNN-based models for skin detection, but they also highlight limitations in handling diverse lighting conditions and varying skin tones.

3. DEEP LEARNING IN FASHION RECOMMENDATIONS

Deep learning models have been effective in processing large amounts of image and text data to improve fashion recommendation accuracy. Sivaranjani et al. (2023) used CNNs, KNN, and RESNET50 to classify and recommend fashion items based on visual features. However, these models are often limited by data imbalance issues and the inability to account for non-visual preferences. The study by Wazarkar et al. (2022) also illustrated how deep learning models can categorize body types, but their dependency on vast datasets poses scalability challenges.

4. AUTO ML AND HYBRID MODELS:

More recent research, such as Yin et al. (2023), explores Auto ML techniques to enhance deep recommender systems by optimizing model components and balancing exploration with exploitation. However, while Auto ML shows promise, it struggles to streamline system components effectively and adapt to rapidly changing trends without continuous model refinement. The proposed Fashion Advisor Model builds upon these studies by addressing the limitations identified in existing systems. By incorporating collaborative and content-based filtering with advanced skin tone and

body type classification, it seeks to overcome challenges related to data requirements and personalization accuracy. The integration of GANs, Decision Trees, and Random Forests further enhances the model's adaptability, aiming to deliver a more inclusive and user-centric recommendation system.

III. LITERATURE REVIEW

Personalized fashion recommendation systems have seen significant advancements, focusing on tailoring suggestions to user preferences, body types, and skin tones. Kuncham et al. (2024) and Vinitha et al. (2024) employed collaborative and content-based filtering, highlighting their efficiency in generating recommendations. However, these systems face limitations such as the cold-start problem and the need for extensive datasets to achieve accuracy. Body type classification has been a focal point in several studies. Rode et al. (2024) utilized machine learning algorithms to analyze body shapes, while Wazarkar et al. (2022) leveraged deep learning for similar purposes. Despite these efforts, challenges in obtaining accurate body type data and dependence on substantial training datasets remain persistent obstacles. Skin tone detection has also been explored to enhance inclusivity in fashion recommendations. Chakraborty et al. (2021) and Daithankar et al. (2014) examined the efficacy of color models such as RGB and HSV for skin tone analysis. While CNN-based approaches improved accuracy, they struggled with lighting variations and the diverse range of skin tones. Deep learning methods have enriched fashion recommendation systems by processing visual features and text data. For instance, Sivaranjani et al. (2023) employed CNNs, KNN, and RESNET50 to recommend items based on image analysis, while Yin et al. (2023) explored Auto ML techniques to enhance model components. These methods face scalability challenges and issues with imbalanced datasets, limiting their broader applicability. Hybrid approaches integrating advanced algorithms like GANs and filtering methods have shown promise in creating user-centric recommendation systems. Tu and Dong (2010) pioneered early models combining multimedia mining with color analysis, while Iso and Shimizu (2021) incorporated personal style analysis for more nuanced recommendations. However, adapting to rapidly evolving fashion trends remains a significant hurdle. The proposed Fashion Advisor Model builds upon these foundations by addressing limitations in existing systems. By integrating advanced algorithms such as GANs, Decision Trees, and CNNs, and considering cultural preferences, the model aspires to deliver highly personalized and inclusive recommendations. Its focus on real-time adaptability and user feedback mechanisms marks a critical step toward improving accuracy and satisfaction in fashion advisement systems.

IV. METHODOLOGY

1. Data Collection and Input Mechanism

Data will be gathered through both manual entry and image recognition methods. Users will select their body type from predetermined options, such as Rectangle, Inverted Triangle, Hourglass, Apple, Pear, Spoon, and Diamond. Image recognition will use CNNs to enhance body type detection accuracy and incorporate user-uploaded photos. The HSV color model will support a detailed classification of skin tone to guide recommendations for the most complementary colors.

2. Algorithms and Model Architecture

The recommendation model will leverage a combination of machine learning techniques and deep learning algorithms:

- **GANs (Generative Adversarial Networks):** Employed for generating diverse outfit combinations based on body type and skin tone, GANs will enable the creation of visually appealing outfit options.
- **Content-Based and Collaborative Filtering:** Content-based filtering will analyze clothing features that match user attributes, while collaborative filtering will consider peer preferences, enhancing recommendations through combined insights.
- **K-Nearest Neighbors (KNN):** Used for locating fashion items with similar attributes, KNN will further support accurate and efficient recommendations.
- **Convolutional Neural Networks (CNNs):** CNNs will be critical for both skin tone classification and body type verification, using uploaded images to better detect and suggest suitable styles.
- **Decision Trees and Random Forests:** To increase the model's robustness, Decision Trees and Random Forests will aid in determining the best classification patterns for different body types, based on user feedback and model training.

3. User Feedback Loop

User feedback will be collected through images and detailed comments, allowing the model to refine recommendations over time. By continuously learning from user interactions, the model adapts to evolving preferences and changing trends. Feedback will also help address the cold-start problem by allowing the model to learn from each unique interaction.

4. Web-Based Interface Development

A Flask-based interface supported by HTML, CSS, and JavaScript will enable users to input their data and interact with recommendations intuitively. Users will be able to manually select their body type, upload images, and view generated outfit combinations. Skin

tone classification through HSV will be executed directly within this interface, enhancing processing speed and user experience.

V. RESULTS AND ANALYSIS

1. Model Evaluation

To validate model performance, the Fashion Advisor Model will be tested on an image dataset covering various body types and skin tones. Key metrics such as accuracy in body type and skin tone detection, as well as user satisfaction based on feedback, will be used to evaluate the effectiveness of each algorithm. GAN-generated outfit combinations will be evaluated based on visual quality and user acceptance rates.

2. Comparative Analysis

The model will be compared to existing recommendation systems lacking skin tone and body type personalization. Traditional methods that focus on user purchase history alone are expected to underperform in terms of relevance and user satisfaction, especially within culturally rich demographics. This comparison will highlight the effectiveness of combining GANs with machine learning algorithms for enhanced personalization.

3. Discussion

The Fashion Advisor Model demonstrates potential for increasing user satisfaction by offering recommendations that respect and celebrate individual body types and skin tones. Integrating cultural attire considerations adds a distinct advantage in markets like India, where fashion preferences are strongly influenced by traditional attire and evolving modern styles. By considering regional attire, the model aligns with cultural sensitivities and enhances users' confidence in their clothing choices. This approach challenges existing fashion recommendation systems, which often miss important details by relying solely on general attributes. The GAN-generated combinations and diverse filtering techniques elevate the model's capacity for personalization, showcasing AI's ability to adapt to highly individualized fashion requirements.

4. Conclusion and Future Work

The proposed Fashion Advisor Model addresses a critical gap in the current fashion recommendation landscape by providing culturally relevant, body-type, and skin-tone specific recommendations. Based on AI and machine intelligence, the idea facilitates decision-making for customers in designer studios while promoting individual expression. Future improvements will include real-time trend monitoring and expand the model to include more body types and outfit combinations to ensure that suggestions align with current fashion trends. By enhancing personalization, this approach encourages inclusivity

and self- assurance in users, making purchasing more enjoyable and personalized. The model's success in the Indian market might set a benchmark for fashion advising systems elsewhere, particularly in regions with similar degrees of cultural diversity and a need for tailored guidance

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“AI POWERED LEGAL DECISION MAKING ASSISTANT SYSTEM”

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Abstract - The integration of Artificial Intelligence(AI) in the field of law is transforming conventional practices, providing greater efficiency, accuracy and accessibility. The paper illustrates the design and development of an AI-based Legal Decision-Making Assistant System. It is made to support legal professionals , judges and policymakers in making well-researched and consistent decisions. The system uses Natural Language Processing(NLP), Machine Learning (ML) and legal knowledge graphs to process massive amounts of legal texts, precedents , legislation and case law. Through human-like reasoning and understanding simulated by the system, the assistant can deliver case-specific results, predict possible outcomes , suggest relevant legal rules and detect meaningful patterns. The solution not only speeds up legal research but also offers transparency and fairness in judicial decision -making .Ethical safeguards and explainable AI(XAI) principles are incorporated in the system to ensure accountability and trust. This research illustrates how AI can revolutionize legal reasoning and provides directions for future research to incorporate it into judicial and legal advisory systems.

I. INTRODUCTION

The judicial process plays critical role in the satisfaction of justice, quality in society. Nevertheless increasing , dependency in cases , legal complexity in proceeding , labour intensive manual research have resulted in efficiencies the judicial decision making process.

According to the most recent reports courts in different jurisdictions are seriously backed up, stretching case hearing delays and exerting tremendous pressure on legal research take considerable time, labour , resulting in disparities and human fallibility. With the emergence of Artificial Intelligence (AI) and Machine Learning (ML) legal profession is also changing . AI – based Legal Decision Making-Assistant System are being developed to assist lawyers, judges and legal experts by automating legal research , case analysis, and judgement prediction.

These systems employ Natural Language Processing (NLP),DeepLearning and knowledge Graphs to process vast legal data, find applicable Case laws and provide data-driven insights to improve decision making. This study presents the development and design of an AI- based legal assistant system that improves legal efficiency through case analysis automation, classify of similar legal case and predictions of probable outcomes based on historical data. The study also explores the potential of AI in minimizing human bias, making the law more accessible and generally enhancing the effectiveness of the judiciary. While ,AI provides promising ,solutions ,ethical concerns such as bias in AI models and accountability in legal decision making are issues that require serious attention.

The goal of this research is to critically assess current AI, models used in legal analytics ,purpose a

framework for an AI – based legal assistant and evaluate its effectiveness in actual legal applications. Through the incorporation of AI in the legal field, this research aims to contribute to the development of a more effective, data-driven and accessible legal system.

II. BACKGROUND & RELATED WORK

The legal profession is going through a revolutionary change with the emergence of Artificial Intelligence (AI) technologies.Traditionally ,legal decision-making has relied upon human interpretation of statutes,judicial precedents ,and facts of cases an exercise time -consuming in nature based on expertise and critical analysis.

The exponential growth of legal data,complexity of cases and requirements of uniform and timely decisions ,however ,has made intelligent systems with the ability to support decision making processes a necessity.AI-based legal decision -making support systems are a rapidly developing field ,driven by the promise of significantly enhancing efficiency and access to justice. Legal professionals have to deal with a huge amount of information , such as case law, statues, regulations and contracts.AI can process and analyze such information more effectively . NLP plays a key role in analyzing legal texts, extracting valuable information and comprehending legal concepts.ML algorithms can be employed to forecast legal outcomes , legal data trends and legal risks. Legal chatbots can assist in providing simple legal information and responding to frequent questions.

This can enhance access to legal aid and reduce the workload of legal professionals. Employing Bert models in legal decision making assistant system to load pre-trained Bert based Question answering model. An improved user interface would be needed for practical use.

III. PROPOSED SYSTEM

Here, we will demonstrate a tool that is designed to assist lawyers, judges, and researchers in analyzing legal cases, summarizing key information, and providing insights based on facts in bid to assist in making informed decisions. It is a natural language chatbot that communicates with users via a Question-Answer(QA) interface. The chatbot has been trained using a well-annotated legal dataset(for instance, constitutional law).

Get the correct legal rules, court cases, and explanations. Offer initial legal opinions or advice based on available knowledge .Have human -like conversation with a natural language interface.

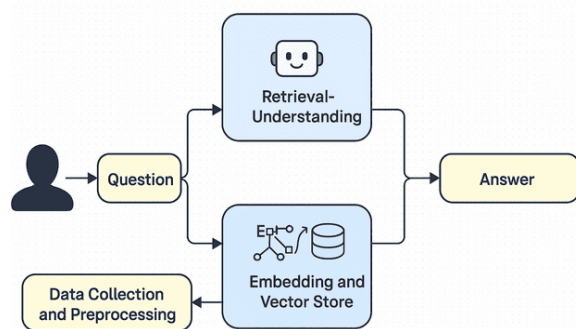


Fig 1. Proposed Method Framework

Figure 1 shows the proposed framework for this study. A brief introduction about system components are as follows:-

1. **Data Collection and Preparation :** Legal texts (Indian Constitutional law dataset) are gathered and cleaned. The data is converted to a structured form appropriate for indexing and retrieval .
2. **Data preprocessing :**Tokenize the question and context using Bert Tokenizer Fast. Offset Mapping align character based answer start with token based offsets. Dataset Splitting 80:20 ratio for training and evaluation sets.
3. **Model Selection :** Using the pre-trained models like Bert-based-uncased models. Better performance with less labelled data. Predicts start and end positions of the answer span within the context. Can be fine tuned to understand domain specific terms and structure. Fine tuning enables the model to adapt to legal domain specific language ,making it robust automated legal decision support.
4. **Fine Tuning the Model:** Fine-tuning was conducted using the Hugging Face Trainer API, which provided an abstraction layer over the training loop, optimizer and scheduler. Training was performed with a batch size of 2 to accommodate ,while gradient accumulations steps were employed to stimulate a larger effective batch size. A single epoch was used for

prototyping ,with the loss function managed internally by the Trainer using cross -entropy for start and end position classification.

5. **Evaluation and Metrics:** In these project we are using evaluation metrics SQUAD v1.1 metric. F1 score also used to measures or overlap between prediction and ground truth.
6. **Model Saving and Deployment:** Save fine -tuned model and tokenizer locally using Save Pretrained ().Integrates with the Gradio for creating the simple and creative UI for chatbot.

IV. APPROACHES / METHODOLOGY

A. Legal Data Collection

Ensures the constitution ,statues, case law ,legal commentaries, court judgements sources. Ensures the information given at a Question Answer Format.

B. Preprocessing and Annotation

Clean unwanted characters, format case names, footnotes removal .

Breaks large text into semantically significant chunks(clauses ,sections).

C. Natural Language Processing(NLP)

Grasps the relevant text and retains tokens and identifies names entities, such as people organizations ,locations , dates andothers , in unstructured text.

Comprehends clause structure and context of legal sentences . Translates user queries and legal content into high dimensional vectors.

D. Legal Domain Adaptation

Tune the base models (e.g., BERT,T5) with domain -specific training datasets such as Indian legal Corpus, Constitutional QA pairs.

Enhance the context interpretation of legal terminology.

E. Answer Generation

Answer generation is the last and most important step in a legal QA system. Once retrieving the most suitable legal content, the system must generate a consistent ,contextually correct, and legally valid response that users will rely on .Utilizes pre-defined patterns to reply to predictable questions(articles).

Utilizes a language model(e.g.,GPT-4,T5,BART) to conditionally generate answers on both the question of the user and retrieval legal context.

V. RESULTS & ANALYSIS

- 1) The performance of the AI based legal assistant system was measured both in terms of quantitative measures(e.g., reviews from legal experts, customers satisfactions

Feature	Description	Result
Legal QA Handling	Ability to answer direct legal questions.	65%
Evaluation Metrics	SQUAD Style QA Exact Match, F1 Score	65% 58%

- 2) Qualitative Evaluation Metrics ,SQUAD style QA and Exact Match.
- 3) The system correctly identified the article text , and explained that Article 21 includes the right to life, dignity ,privacy ,and legal aid.
- 4) The chatbot successfully tracked the context and provided an appropriate , cited answer.
- 5) Below the graph shows a comparison between Right Case law and Procedural Law + Definitions.

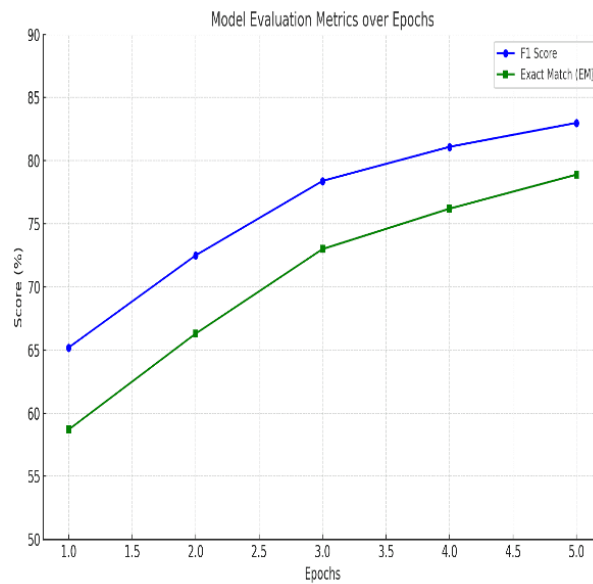


Fig 2. Accuracy Metrics Graph.

TECHNIQUES

1) Natural Language Processing(NLP)

Legal Text Parsing & Preprocessing

Apply tokenization ,sentence segmentation , named entity recognition (NER), and part- of -speech tagging specific to legal texts.

Legal Document Understanding

Utilize models such as Legal Bert, Case Law Bert ,or fine-tuned LLMs to analyze contracts, statutes , and case law. Provide brief summaries of voluminous legal documents or judgements. Apply retrieval - augmented QA systems for legal statutes and case-based queries.

2) Transfer Learning

Transfer learning is a foundation of deep learning technique that can use pre-trained models for decision making assistant it can plays major role in pre trained

Bert-base-uncased model. We fine tuned this pre-trained model on a domain specific, SQUAD style legal QA dataset consisting of Indian constitutional texts and their corresponding question answer pairs. Fine tuning enables the model to adapt its parameters to learn domain specific answer span detection in legal documents, while preserving its general understanding of syntax, semantics.

3) Supervised learning with fine-tuning.

Supervised learning with fine-tuning is a specialized form of machine learning where a pre-trained model is adapted to a new task using a labelled dataset. The supervised fine tuning approach significantly reduces the data and computer resources required for high accuracy. The use of Hugging Face’s Tainer API allows efficient batching ,gradient accumulation and epoch based training with real time evaluation using SQUAD metrics.

4) Model Evaluation

Model evaluation is important phase in developing NLP systems to ensure the model performs accurately, reliable and generalizes well to unseen

data. In our AI powered legal decision making assistant system, we evaluate the fine tuned Bert based question answering model using standard extractive QA metrics: Exact Match(EM) and F1 score, based on the SQUAD evaluation framework. The EM metric measures the percentage of predictions that exactly match the truth answers

,while the F1 score captures partial overlaps by computing the harmonic mean of precision and recall over predicted and true answer spans.

5) Model Serving

Using the frontend for chatbot of Gradio. Creating simple attractive UI interface .

LITERATURE SURVEY

Sr No.	Title	Author's Name	Year	Methodology /Algorithm used	Accuracy Achieved
1.	Artificial Intelligence in law making law enforcement :Risk and new opportunities.	D.A. Pashentsevl, Yu.G.Babaeva	2024	Support Vector Machines(SVM) Natural Language Processing(NLP)	Model Accuracy:89.3
2.	AI -Driven Smart Management Processes: Transforming and Decision Making and shaping the future.	Ankita Shukla,2Dr.Alka Agnihotri.	2024	AI integration in management practices such as decision making processes and process automations. Algorithms used such as Machine learning(General), Neural networks, Natural Language Processing(NLP), Cognitive AI systems.	R2 Score:30.7% DetectionAccuracy:95% Accuracy:90
3.	AI Driven Justice: Evaluating the Impact of the Artificial On legal systems.	Rachid Ejjami	2024	Methodology used problem formulation defining the scope. Algorithms :Natural Language Processing. AI Driven Decision Support Systems.	Article does not present empirical data or specific accuracy metrics for AI applications in legal systems.
4.	AI Powered legal documentation Assistant.	P.VimalaImogen.J Sreenidhi,V.Nivedha	2024	1. NLP 2. ML 3. Deep Learning(DL) 4. Rule based + Template Matching.	1. ANN - 94.02% 2. SVM - 85.65% 3. DT - 81.68%
5.	How Does Artificial Intelligence Improve decision making Evidence from the AI powered Go program.	Sukwoong choi Hyo kang NamilkingJunsik Kim	2023	1. Artificial Neural Network. 2. Decision Tree 3. Naive Bayes	The result show significant improvements in the efficiency and accuracy of document preparation.

6.	Exploring the Ethical Implications of AI in Legal Decision Making.	Raj Kumar Sharma	2023	1.Machine learning for case outcome prediction. Natural Language Processing(NLP)for legal document analysis.	1. GA 84.03% 2. SVM 94.8%	- -
7.	Navigating the Maze:AI and Automated Decision Making Systems in Private International Law.	Mukhammad Ali Turdialiev Tashkent State University of Law.	202	1. K-Nearest Neighbours (K-NN) 2. Cluster center and Nearest Neighbours (CANN) 3. SVM	1. CANN 99.76% 2. KNN 93.87% 3. SVM 80.65%	- - -

CHALLENGES

Making a successful AI Powered legal Decision making Assistant System implies a number of major challenges:

1) Complexity and Ambiguity in legal Language.

Constitutional language is vague, wordy , and fraught with subtle legal .

NLP models can error in interpreting legal concepts or they can oversimplify them. Sample :Words and phrases like “reasonable restrictions” or “due process” can be very context dependent as well as their interpretation .

2) Data Quality & Availability of Data.

Data in the form of laws tend to be diverse (cases, statues, agreements), unstructured, and country - specific.

It challenging to access current and complete datasets(case law, regulations) because of licensing or paywalls.

3) Interpretability of AI Models

Deep learning models (particularly LLMs) can give correct outputs , but it is challenging to explain how decisions are reached.

4) Contextual Reasoning

Legal judgements tend to rely on knowledge of precedents , jurisdictional principles ,intent , and factual distinctions.

5) Integration with Existing Legal Workflows

Law firms or courts might have legacy systems that are difficult to integrate with contemporary AI architectures(PDFs, proprietary tools, etc.)

CONCLUSION

In this project we propose that an AI -Based legal decision making assistant system has a vast potential to transform the legal system by making it more efficient, accessible to justice , and enhancing decision making for legal professionals and institutions. With powerful AI models ,such systems are capable of doing routine tasks, handling large

legal databases, and assisting in sound legal reasoning.

But this transition doesn’t take place without significant challenges. The initiative must navigate very carefully through the prisms of legal language ,dataquality ,differences in jurisdiction ,and the ethical mandate of AI deployment in high -risk environments. Trust ,transparency ,and human agency in decision-making are just as important.

Multidisciplinary collaboration is necessary when assembling a team of experts with law, artificial intelligence ethics ,and software engineering backgrounds in order to make the project a success. Ongoing verification of user input , and compliance with regulations should be the guiding force for the development process.

In conclusion ,while path to building an AI legal assistant is challenging ,it is a promising and worthwhile endeavour. If implemented responsibly , such systems can significantly contribute to more accessible and data driven systems.

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At last, I express my deepest gratitude to my family and friends for their assistance and motivation

throughout the duration of this project. Their words of strength and patience were very helpful in staying dedicated and within the limits of our objectives.

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A CASE STUDY OF SYNTHESIS IN VLSI DESIGN USING FUSION COMPILER

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Abstract - Three main constraints—area, timing, and power consumption—have a major impact on the design flow of Application Specific Integrated Circuits (ASICs) in the field of Very Large Scale Integration (VLSI) design. Addressing these limitations becomes more crucial as these circuits' complexity and performance requirements increase. Electronic Design Automation (EDA) tools, which facilitate several stages of the design process, have been developed to manage this increasing complexity. Logic synthesis is a critical step that converts highlevel Register Transfer Level (RTL) code into an efficient gatelevel netlist. To guarantee that the finished implementation satisfies predetermined performance criteria, this transformation takes design limitations into account. Other factors are introduced when Design for Testability (DFT) methodologies are used during synthesis. Although DFT makes the design more testable, it can also result in higher power and area usage, which could affect performance as a whole. It is crucial to perform performance analysis, which includes comparing the synthesis outcomes under different constraint conditions and with and without DFT insertion. These studies aid in comprehending the trade-offs and in making well-informed judgments to successfully optimize the design. Netlists are legally validated against RTL for Logic Equivalence Checking (LEC) using the Conformal EC tool. A Synopsys EDA tool called Fusion Compiler (FC) offers a convenient way to combine synthesis approaches, reducing design cycles and enhancing design quality.

Keywords - VLSI, Synthesis, Logic Equivalence Check(LEC), Fusion Compiler(FC).

I. INTRODUCTION

EDA tools are essential for the design, verification, and debugging of bigger digital circuits because of the quick improvements in VLSI design level of integration during the past few decades. Designing a range of electronic chips using integrated circuits is made possible by EDA tools. The scope for EDA tools is presented by the need to satisfy the functional and performance goals within a design time frame and associated cost. One such EDA tool is the Synopsys Fusion Compiler (FC), which combines user-defined gate arrays and standard cell libraries to carry out the logic synthesis of intricate digital circuits to create quick and space-efficient ASIC designs. The Fusion compiler optimizes both sequential and combinatorial designs and supports a broad variety of flat and hierarchical design patterns. Under varied loads, temperature, and voltage circumstances, it also examines design trade-offs involving design restrictions like timing, area, and power. Additionally, EDA tools are lowering manufacturing errors, cutting manufacturing costs, optimizing IC design, and making complicated ICs easier to use. Meeting the targeted temporal and area goals can improve performance while producing high-quality solutions. However, this presents a difficulty for the logic synthesis of hierarchical modular design utilizing the FC tool. Given a standard-cell library and design restrictions, logic synthesis is the process of translating a high-level design description into an optimal gate-level representation. It is a highly automated process that connects physical design automation and high-level synthesis. The logical equivalency between the RTL

and netlist is verified using the FEV approach. It is now a crucial component of the design process and is used for hand editing later on, such as Engineering Change Orders (ECO). It is a comprehensive verification technique that uses no test vectors to fully confirm design functioning. The powerful and effective instrument for logical equivalency is the Cadence Conformal Encounter Equivalence Checker. The paper is structured as follows: 1. Creating an RTL model for the case study 2. Use the FC tool and perform Physical Synthesis. 3. Optimize the design for best PPA. 4. Using a conformal tool, perform LEC of the generated design. The proposed work's scope is that new synthesis and optimization strategies that yield superior synthesis results are made possible by the use of EDA tools such as design compilers.

II. DESIGN FLOW

RTL modeling is the first step in the design flow employed in the case study, and LEC is the last step. Figure 1 provides an illustration of the design flow description. This process is divided into many sub-steps, which include RTL Design, Verification, synthesis, LEC. Figure 1 depicts the steps involved in the integrated circuit (IC) design flow.

Several essential documents are crucial in running this flow.

- a) **RTL File (all.v):** The files provide a high-level explanation of the digital circuit's functionality and behavior. They specify how the circuit should behave and function.
- b) **Technology Library Files (.lib):** The timing, power, and area related to the standard cells are

- represented in ASCII by the timing library (.lib). The output capacitance (Load) and input transition (Slew) are used to calculate the delay.
- c) **SDC:** The timing and power-related limitations that regulate design in relation to the specification are included in standard design constraints, also known as Synopsys design constraints.
 - d) **UPF:** An IEEE standard called the Unified Power Format (.upf) is used to specify power and associated elements of multi-voltage design. Definitions of supply sets, power domains, power switches, retention cells, level shifter cells, and other low power-related terms are included in UPF.
 - e) **lef file:** The standard cells, macros, and other design components needed for chip layout are described in depth. In addition to metal layer definitions and routing track specifications, it contains details about the physical dimensions of cells, including height, width, and pin locations. The design guidelines for cell placement, routing restrictions, and via configurations are specified in the LEF file.
 - f) **def file:** It records the specific physical layout details of an integrated circuit, such as the positioning of I/O pins, standard cells, and macros, as well as the interconnect routing. It gives the coordinates for every cell and macro on the chip, as well as the positions and orientations of every component. The precise routes of the interconnects as well as details on the metal layers and routing vias are contained in the DEF file. In order to guarantee that the design complies with the given design guidelines and limitations, it also specifies the positioning of blockages and routing impediments.

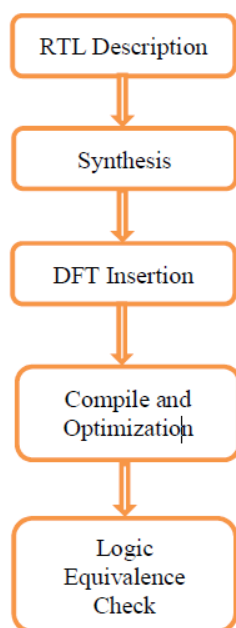


Figure.1: Design flow for case study

A. LOGIC SYNTHESIS

Given a standard-cell library and design restrictions, logic synthesis is the process of translating a high-level design description into an optimal gate-level representation. It is a highly automated process that connects physical design automation and high-level synthesis. Logic synthesis encompasses a broad range of digital system design, with synthesis approaches primarily focusing on synchronous behavior and implementation. The market-driven choices that impact the design philosophies used for product implementation have a significant impact on the quantity of design automation and logic synthesis work. The RTL description in Verilog is first converted by logic synthesis into components taken from the DesignWare and Generic Technology (GTECH) libraries, which are independent of technology. After that, it was mapped to technology-dependent library cells and optimized using the appropriate optimization rules and methods. Architectural optimization, logic level optimization, and gate level optimization are the three layers of optimization that can be carried out during the synthesis process.

B. PHYSICAL SYNTHESIS

Following logic synthesis, an early design floorplan is created for placement data, and based on this floorplan, routing requirements are estimated. This information is provided as an input in .def file to physical synthesis. It captures the detailed physical layout information of an integrated circuit, including the standard cells, macros, and I/O pins placement, as well as the routing of interconnects. We are restricting the tool to place within a certain area. Real Physical Synthesis is more accurate, but runtime is more.

C. DFT Insertion

These days, DFT is crucial to the design flow of VLSI ASICs since it makes it possible to include testability characteristics in the design stage itself. It guarantees that every flaw in the design will be found. Adding additional signals and logic to a test in the design is the underlying idea of structured DFT, which is simple to implement. SCAN and BIST are the two structured approaches that are most frequently utilized. Many sequential elements in the design can be made more controllable and observable by using the SCAN Shift and Scan Capture approach. The most well-liked and extensively applied structured DFT method is SCAN Shift and Scan Capture.

To make the process of creating test patterns easier, the scan approach converts all sequential elements into combinational elements. By joining all of the flip flops in series to create a shift register known as Scan Chains (SC), the scan technique aims to incorporate the test mode logic into a circuit. It links scan inputs, or PIs, and scan outputs, or POs.

All of the original conventional flip flops in the design can be swapped out for scan flip flops (SFFs) to create a scan chain. Three different kinds of scan flip flops, including LSSD, multiplexed, and clocked forms, can be employed in scan insertion. Multiplexed scan cells are the most popular flip flop in scan design because they work well with contemporary designs and are fully supported by automation technologies now in use. A flip flop's input is coupled to two more input multiplexers in a multiplexed SFF.

D. Logic Equivalence Check

The logic equivalency between the netlist and RTL is verified using LEC. It is now a crucial component of the design process and is used for hand editing later on, such as Engineering Change Orders (ECO). It is a comprehensive verification technique that uses no test vectors to fully confirm design functioning. If both models are logically identical, LEC is a very quick and simple procedure that essentially confirms the rapid modifications to the model designs.

The golden model is the name given to the RTL design paradigm at the behavioral abstraction level. This is combined to create the netlist, which is regarded as the updated model at the gate level of abstraction. There are two LEC modes: setup and LEC. The RTL design's constraints are specified by the setup mode. The LEC technique uses the mapping, comparison, and diagnosis procedures to validate the RTL design in LEC mode. In the LEC mode, it first maps each significant design point. The necessary parts include tie z, tie gates, black boxes, cut gates, D latches, D flip flops, primary inputs, and primary outputs. Combinational logic cones enclose the design's primary points. The cones of logic must line up in order to compare two designs using the key point mapping process.

- Mapping

For later comparison, mapping compares important elements of the original design with their equivalents in the modified design. Function-based mapping and name-based mapping are the two categories of key point mapping. Module and signal names are mapped in designs using name-based mapping. It takes longer to map logic functions like gates, flops, and primary inputs in models using function-based mapping than name-based mapping.

- Compare

Use the comparison command in LEC mode after mapping important locations with the conformal tool. Black boxes, D flip-flops, D latches, and principal outputs are examples of comparison points that designers can manually define.

- Analyze

The method looks for possible faults by examining non-equivalent points. The conformal tool shows the underlying error patterns and detects non-equivalent spots. To determine the fan-in or fan-out cones of

non-equivalents, the tool employs gate reporting. Visual accessibility is enhanced by the schematic representation and color coding of error candidates, nonequivalent points, diagnostic input points, and supporting critical points.

III. IMPLEMENTATION

Verilog HDL is utilized to explain the case study's RTL design. Writing the RTL verilog code in a way that facilitates synthesis is essential. The Fusion Compiler tool accepts the case study's RTL verilog code as input. Following the HDL to gate translation, the FC tool optimizes the design and maps it to a particular technology library known as Target Library. This process is guided by constraints. Details on cell characteristics, features, and design operating conditions are included in the target library. Using the symbol library, the synthesis tool creates a gate level schematic and an RTL design schematic. To resolve references, the fc tool makes use of both the link and target libraries. All libraries with cells referred by the netlist are listed in the link library. Time and path delays are computed using delay models defined by the link library. The designer has additional control over how the cells are mapped thanks to the target and link libraries. They can also remap a gate-level netlist between different technologies.

The analysis command transforms HDL designs into an intermediate format in the work directory after examining them for syntax and synthesizable logic. Elaborate then takes this as input and starts working on turning the RTL description into physical hardware. The processes of analysis and elaboration quickly determine whether the design intent of the tools is being met. During synthesis, the check_design command is used to fix design flaws and verify the internal representation. Unconnected ports, constant valued ports, recursive hierarchy definitions, connection class violations, cell and reference mismatches, numerous driver nets, and cells lacking input or output pins are among the errors that occur during check_design.

The FC tool reads optimization constraints as input. The Fusion compiler optimizes designs using design rule restrictions and optimization constraints during synthesis. Technology libraries define implicit design rules. These requirements are necessary for a design to work properly and can be stricter than the optimization constraints. It includes the maximum fan-out, the minimum and maximum capacitances, cell degradation, and the maximum transition time. These properties cannot be removed because they are essential for technology, however more limited values can be set.

The design's speed and area objectives are represented by the optimization constraints. Area is not as crucial as speed. Design rule constraints take precedence over these limits. Timing limitations for synchronous pathways include input and output

delays, while asynchronous paths have maximum and minimum delays. Timing for critical paths, false paths, and multi-cycle paths should be carefully considered. The `gui_start` command loads the FC tool's design vision, which allows RTL and netlist schematics to be shown. It provides dialogue boxes for commands to the FC. It also aids in analyzing the design at the GTECH and gate levels.

The LEC tool has two modes: setup and LEC mode. To use the LEC tool in setup mode, designers must input the golden RTL and revised gate level designs, as well as the standard libraries needed for DC synthesis. The design attributes and constraints are also defined, which are required for the LEC tool to perform equivalency checks on the designs.

The designer can define comparison points between golden and amended designs. The DFF and POs are defined as comparison points. If the compare points are comparable, the RTL and netlist designs are logically equivalent. To diagnose unmapped or non-equivalent sites, use tools such as the diagnosis

manager, schematic tracer, source code manager, and gate manager.

IV. RESULTS AND DISCUSSIONS

1. Timing Report

After executing the synthesis, we observed setup violations in the design. Achieving timing clean results in a design that meets all timing specifications, ensuring functional correctness and reliability.

Type	Path Group	NUM	WNS	TNS	Status
Setup	Reg2Reg	23	-31	-94	Dirty

Table.1: Timing Report after the Synthesis execution

NUM – No of Violating Paths

WNS – worst negative Slack

TNS – total negative Slack

Figure 3 illustrates the flow of formal equivalence verification using the cadence conformal tool.

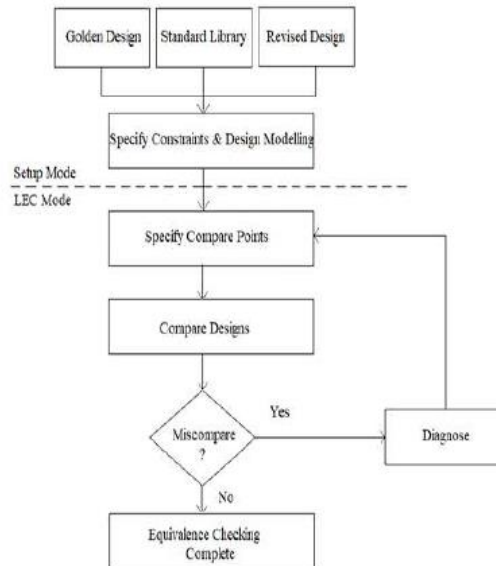


Figure 3: LEC flow diagram of Conformal EC tool

2. Strategy used for fixing Timing Violation

In the first step initially focused on the critical paths which are causing setup violations. Generally setup violations occur when your data path is slower. So I made data path faster by including ULVT cells, for which delay will be less.

Type	Path Group	NUM	WNS	TNS	Status
Setup	Reg2Reg	6	-5.41	-32.20	Dirty

Table.2: Timing Report after first iteration

After this power has been increased because of ULVT cells. In the second iteration I have increased the drive strength of the cells in the critical paths and reduced the percentage of ULVT cells. After the next iteration there is no new timing violation and the

timing is clean, with reduced power consumption and minimal increase in area.

Type	Path Group	NUM	WNS	TNS	Status
Setup	Reg2Reg	0	0	0	clean

Table.3: Timing Report after second iteration

3. Power Report

Leakage	Internal	Dynamic	Total
1.08+e^5 nw	1.595+e^6 nw	5.82+e^5 nw	2.3+e^6 nw

Table.4: Power Report

After synthesis, the Conformal EC tool is used to run the LEC with inputs including golden and amended

netlist designs, as well as the standard cell library. The LEC run is successful if both designs have an equal number of mappings and compare important points. The golden and updated models each have 18268 critical locations for mapping logical equivalence checks. The LEC does not use PI as a crucial factor to compare both designs. The result of LEC run is summarized in Table 5.

Mapped Pointss	Golden	Revised
PI	2011	2011
PO	1643	1643
DFF	14614	14614
Total	18268	18268

Table.5: Mapped Points of LEC

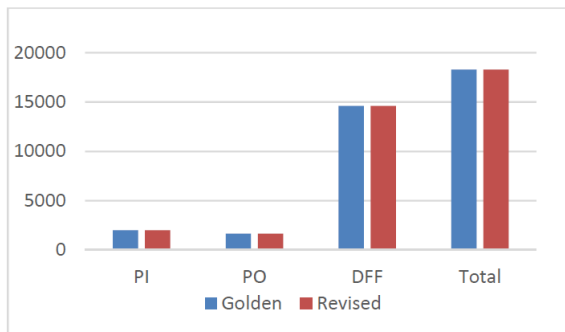


Figure 4: Comparison of LEC

V. CONCLUSIONS

Using EDA tools throughout the design process improves accuracy and quality of outcomes. Designing using hierarchy, splitting, and parallelism reduces operation time. Using EDA tools can save design time and costs. The design compiler uses its builtin mapping and optimization tools to optimize the case study. The example study effectively implements most of the ASIC design flows, including RTL model development and LEC. Our work offers significant insights and practical solutions for VLSI designers, helping to optimize verification processes in the semiconductor industry. Our technique aims to improve the efficiency and reliability of integrated circuits, which are essential for modern electronics.

Future work will focus on further improving the optimization techniques and expand their application to other VLSI blocks and technologies. Our research highlights the significance of novel methodologies in VLSI design, resulting in more dependable and high-performance semiconductor devices.

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PERFORMANCE ANALYSIS OF DOUBLE-GATE STAGGERED HETEROJUNCTION TFET BIOSENSOR

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Abstract - This study introduces a double-gate staggered heterojunction tunnel field-effect transistor (DG-SHJ-TFET) optimized for high-precision biosensing. The design features an InAs heterojunction pocket at the source-channel interface and a nanoscale cavity on the source side, which collectively improve band-to-band tunneling efficiency and overall sensor sensitivity. The study investigates the impact of biomolecule dielectric constants ($\epsilon_k = 1$ to 12) and charge densities (neutral, positive, negative) on device electrostatics, including surface potential, energy band alignment, and drain current characteristics. Results demonstrate that increasing the dielectric constant (ϵ_k) improves sensitivity, with positively charged biomolecules exhibiting the highest response due to enhanced electric field effects. The device achieves 60× higher drain current sensitivity (S_{Id}), almost double the threshold voltage shift (ΔV_{th}), and 12 times increase in ON-OFF current ratio sensitivity ($\Delta(I_{on}/I_{off})$) compared to conventional FET-based biosensors, further optimizing performance. These findings highlight the potential as a highly sensitive, low-power biosensor for detecting biomolecules in medical diagnostics, environmental monitoring, and lab-on-chip applications.

Keywords - DG-SHJ-TFET, Biosensor, Sensitivity

I. INTRODUCTION

The continuous advancement in nanotechnology and biomedical engineering has accelerated the development of miniaturized, low-power biosensors capable of detecting biomolecules with high sensitivity and specificity. Some of the transistor structures include Tunnel Field-Effect Transistors. TFETs have also demonstrated potential in biosensing applications, taking advantage of their low-voltage operation and sub-thermal subthreshold swings.

The traditional MOSFET-based biosensors have the weaknesses of excessive leakage currents and poor sensitivity at the nanoscale size [1-2]. Instead, band-to-band tunnelling (BTBT) is the dominant carrier injection mechanism employed by TFETs, and TFETs rely on it to produce a steep subthreshold slope and very low off-state current. Because of these properties, TFETs are well suited to applications where low power and high sensitivity are required, e.g., handheld diagnostic and integrated lab-on-chip platforms.

Other structural developments such as many gate topologies, and staggered heterojunctions, have also been explored to enhance the performance of TFET biosensors [3]. The staggered heterojunction provides better tunnelling characteristics by tuning the energy band alignment between the source and the channel regions and the stacked gate structure provides better electrostatic control over the channel that results in improved switching characteristics. In this thesis, a new kind of biosensor is studied using a Double Gate Staggered Heterojunction TFET (DG-SHJ-TFET) [4]. A nanogap is created in the gate dielectric area to interact with the target biomolecules. A change in the

dielectric constant or the charge content of the molecule results in effective variation of electrical properties of the sensor. The proposed structure is simulated and analyzed thoroughly, along with discussing its performance with the crucial parameters of biosensing like sensitivity, I_{on}/I_{off} ratio, and subthreshold swing, which demonstrates its suitability in the future biological sensing technologies [5].

II. DEVICE ARCHITECTURE AND SIMULATION SETUP

The biosensor incorporates a double-gate TFET device structure that has a strategically engineered hetero-pocket formed at the source-channel junction that significantly increases the efficacy of band-to-band tunneling (BTBT). The device is designed and manufactured on 100 nm silicon substrate making the best use of its compact structure to provide better performance. One of the key constituents is a 10 nm thick layer of HfO₂ deposited by LPCVD and an aluminum gate etched in dry etching. The selective etching of HfO₂ layer is done to form nanocavities on both sides of the gate close to the source area.

A tiny layer of silicon is exposed by the nanocavities, which immobilizes biomolecules. Charge transport is enhanced by electrodes around the source and drain and minimized leakage. Simulations were carried out using Silvaco TCAD tools, utilizing realistic models like carrier mobility, recombination, and quantum transport. Incorporating a concentration-dependent mobility model, a recombination model used in this like consrh, and a Fermi-Dirac statistics transport model to account for the effects of heavy doping densities. Table I shows the device specification.

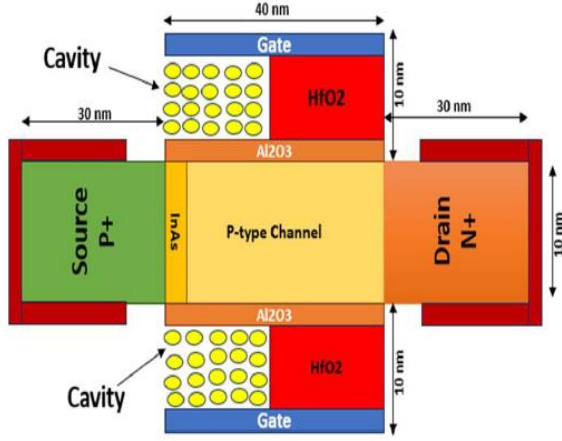


Fig.1 Proposed Device Structure

Parameters	Value
Doping density N^+ (cm^{-3}), ND	1×10^{18}
Doping density P^+ (cm^{-3}), NA	1×10^{20}
Source Length (nm), L_s	30
Source Thickness (nm), L_S	10
Drain Length (nm), L_d	30
Drain Thickness (nm), T_d	10
Gate Length (nm), L_g	40
Gate Thickness (nm), T_g	10
Pocket Length (nm), L_p	5
Pocket Thickness (nm), T_p	10
Length of Nano-cavity (nm), L_c	20
Thickness of Nano-cavity (nm), T_c	5
Gate-Oxide Thickness (nm), T_{ox}	5
Channel Length (nm), L_{ch}	40

Table I: Device specification

Assurance of calibration against experimental results accurate prediction of the biosensor's electrical behavior. This design provides encouraging advantages for label-free detection of biomolecules at the nanoscale [9].

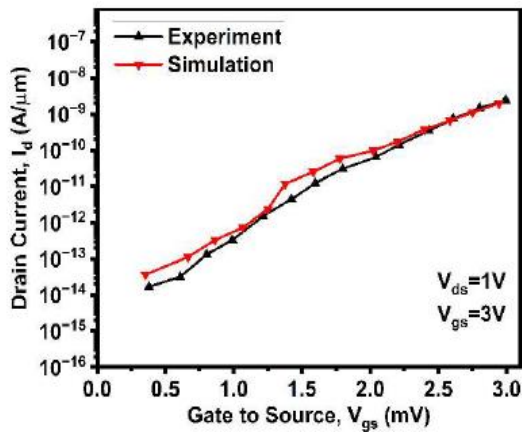
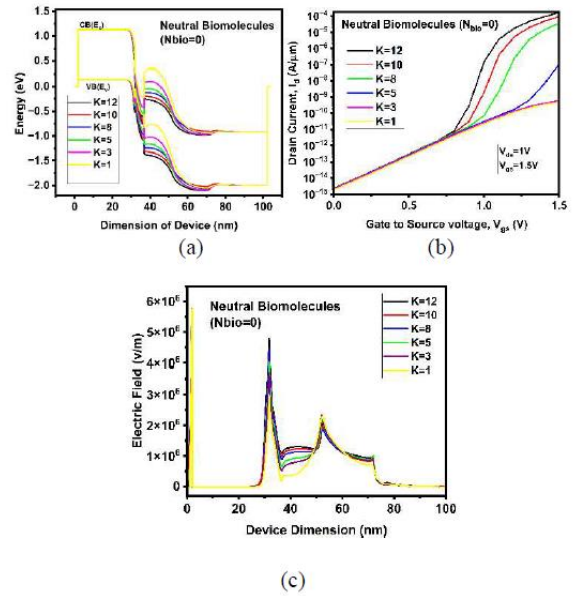


Fig.2 Calibration graph

III. RESULTS AND DISCUSSIONS

A. DC Characteristics

The DC characteristics of the proposed DG-SHJ-TFET biosensor were examined using TCAD simulations to understand its behaviour in various situations. For $\epsilon k = 1$ (air), the device shows minimal sensitivity, but as ϵk increases (3, 5, 8, 10, 12), the drain current sensitivity (SID) improves substantially. These properties describe the connections between the gate voltage that is applied and the consequent drain current and are significant to evaluate the functioning of sensors. Such parameters as energy band diagrams, electric field distribution, tunnelling rate, and I_d - V_{gs} behaviour were studied with emphasis on the role of interface traps, gate oxide properties, and biomolecular charges. [6-7].

Fig.3 DC characteristics for Neutral Biomolecules (a) Energy-Band Diagram, (b) I_d vs V_{gs} Graph, (c) Electric Field Variation

Variations in ON current, subthreshold swing (SS), and threshold voltage were analyzed to understand the sensitivity and accuracy of the biosensor. The implemented PNP configuration in the TFET architecture optimizes charge carrier tunneling through strategic electric field modulation at the source-channel interface [8]. This structural enhancement yields superior ON-state characteristics and achieves a more abrupt subthreshold swing (SS). Furthermore, the inclusion of a heavily doped n-type pocket between source and channel regions generates intensified electric field concentrations at the p-n junction boundary. This dual effect simultaneously amplifies current conduction while effectively mitigating interface trap formation [10].

B. Effect of Positively Charged Biomolecules

For $\epsilon k = 1$ (air), the device shows minimal sensitivity, but as ϵk increases (3, 5, 8, 10, 12), the drain current

sensitivity (SI_d) improves substantially. At $\epsilon k = 12$, the highest sensitivity is observed due to the combined effect of the dielectric constant and positive charge concentration. The positive charges change the electric field and width of the depletion region therefore, the threshold voltage sensitivity (ΔV_{th}) also increases, as the electric field and depletion region width are altered by the positive charges.

These results prove the functionality of the sensor in the precise detection of the presence of positively charged biomolecules. Simulation with positively charged biomolecules showed enhanced channel surface potential, which intensified the electric field at the tunneling junction. A fixed positive charge density of $1 \times 10^{11} \text{ cm}^{-2}$ was introduced for this analysis. The stronger electric field leads to improved band-to-band tunneling, raising the drain current. The band diagram also revealed an upward bending near the channel surface, which enhances the tunneling current. Therefore, the presence of positively charged biomolecules significantly improves the biosensor's sensitivity by increasing the band bending and tunneling efficiency [11-12].

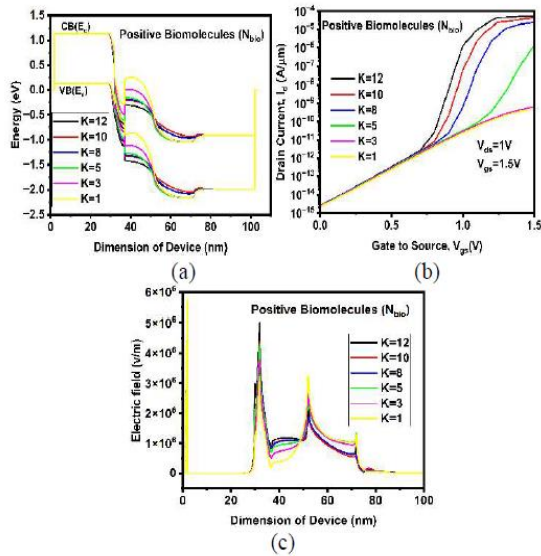


Fig. 4 DC characteristics for Positive Biomolecules (a) Energy-Band Diagram, (b) Id vs Vgs Graph, (c) Electric Field Variation

C. Effect of Negatively Charged Biomolecules

In contrast, negatively charged biomolecules were found to suppress the tunneling process. These charges reduce the surface potential, weakening the electric field at the junction. Consequently, the tunneling probability decreases, leading to a noticeable reduction in drain current [13]. The energy band diagram shifts downward, increasing the barrier height for tunneling. This drop in current response confirms the biosensor's ability to distinguish between biomolecule polarities, with reduced sensitivity in the case of negative charges [14].

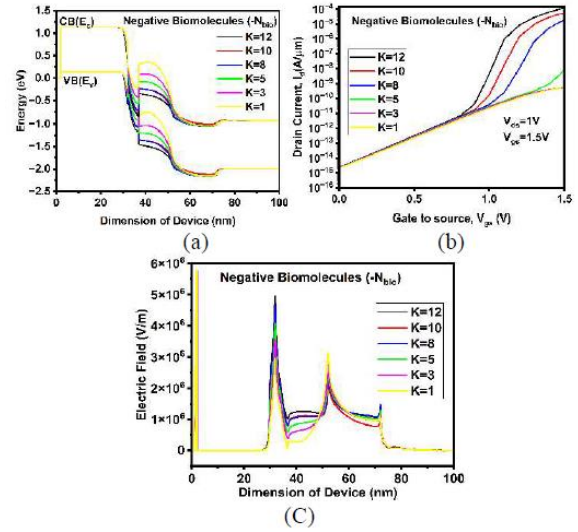


Fig.5 (a) DC characteristics for Negative Biomolecules (a) Energy-Band Diagram, (b) Id vs Vgs Graph, (c) Electric Field Variation

When positively charged biomolecules ($N_{bio} = 1 \times 10^{11} \text{ cm}^{-2}$) are introduced into the nano-cavity, the device exhibits significant improvements in sensitivity [15]. The presence of positive charges elevates the surface potential at the tunnelling interface, intensifying the local electric field magnitude. This field enhancement enhances electron tunneling efficiency, thereby increasing the drain current (I_d). The mechanism occurs through effective gate voltage amplification, which enhances charge carrier accumulation in the conduction channel [16]. The energy band diagram shows an upward shift in the conduction and valence bands, reducing the tunneling barrier and enhancing current flow. In Table II shown biomolecules used in this paper for analysis and comparison.

Biomolecule	Dielectric Constant (K)	Charge at pH 7.0
Biotin	2.63	Slightly Negative
Gluten	5	Neutral
Zein	7	Neutral
Chlorobenzene	9	Neutral
Keratin	10	Neutral/Polar
Gelatin	12	Positive
Glutamic Acid	12	Negative

Table II: Biomolecules and Their Dielectric Constant

IV. SENSITIVITY ANALYSIS

Two significant figures were used to characterize the sensitivity of the DG-SHJ-TFET used to assess the biosensor: drain current sensitivity (SI_d), threshold voltage shift (ΔV_{th}), and ON-OFF current ratio sensitivity ($\Delta(I_{on}/I_{off})$). Such parameters were considered both in neutral and charged biomolecules in different conditions of dielectric constant.

$$\Delta \left(\frac{I_{ON/OFF}|_{\epsilon_k > 1}}{I_{ON/OFF}|_{\epsilon_k = 1}} \right) = \frac{I_{ON/OFF}|_{\epsilon_k > 1} - I_{ON/OFF}|_{\epsilon_k = 1}}{I_{ON/OFF}|_{\epsilon_k = 1}} \quad [1]$$

$$\Delta V_{th}|_{\epsilon_k > 1} = V_{th}|_{\epsilon_k > 1} - V_{th}|_{\epsilon_k = 1} \quad [2]$$

$$S_{I_d}|_{\epsilon_k > 1} = \frac{I_d|_{\epsilon_k > 1} - I_d|_{\epsilon_k = 1}}{I_d|_{\epsilon_k = 1}} \quad [3]$$

A. Sensitivity Analysis for Neutral Bio-Molecules

Neutral biomolecules ($N_{bio} = 0$) do not contribute additional charges but influence the device through dielectric modulation. As ϵ_k increases (1 to 12), the effective gate capacitance changes, modifying the tunneling junction's electric field. The drain current sensitivity (SID) improves with higher ϵ_k , as the dielectric environment enhances carrier tunneling. The threshold voltage sensitivity (ΔV_{th}) also exhibits a steady rise, demonstrating the device's capacity to identify neutral biomolecules by their dielectric characteristics. However, the sensitivity is less than that of charged biomolecules due to the absence of charge-induced band bending [17-18].

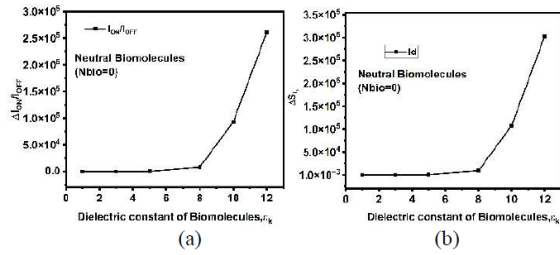


Fig. 6 Sensitivity Analysis for Neutral Biomolecules (a) Ion/Ioff sensitivity, (b) Drain current (Id) sensitivity

B. Sensitivity Analysis for Charged Bio-Molecules

For positively charged biomolecules, the sensitivity metrics (SID, $\Delta(I_{on}/I_{off})$) exhibit significant enhancements with increasing ϵ_k and charge concentration. At $\epsilon_k = 12$ and $N_{bio} = 1 \times 10^{12} \text{ cm}^{-2}$, the drain current sensitivity improves by 60 times, while the threshold voltage sensitivity doubles compared to conventional FET-based biosensors. The ON/OFF current ratio sensitivity increases 12-fold, demonstrating the device's superior performance. In fig.7 shown that the sensitivity analysis of the negative charge biomolecules.

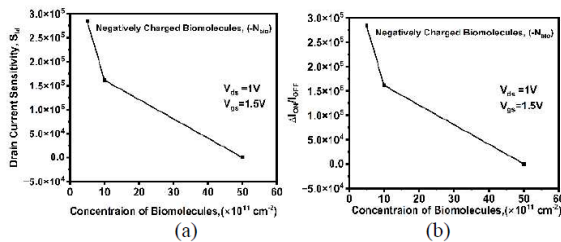


Fig. 7 Sensitivity Analysis for Negative Charge Biomolecules (a) Ion/Ioff sensitivity, (b) Drain current (Id) sensitivity

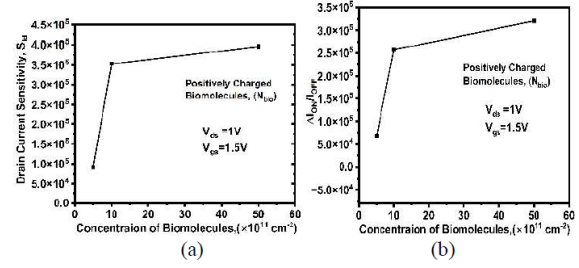


Fig. 8 Sensitivity Analysis for Negative Charge Biomolecules (a) Ion/Ioff sensitivity, (b) Drain current (Id) sensitivity

For biomolecules carrying negative charges, the sensor exhibits reduced sensitivity; however, the response remains detectable. Fig. 8 shown that the sensitivity analysis of positive charge biomolecules. At $\epsilon_k = 12$ and $N_{bio} = -1 \times 10^{11} \text{ cm}^{-2}$, the drain current sensitivity is reduced due to the opposing effect of negative charges on the electric field.

However, the device remains responsive, highlighting its versatility in detecting varying biomolecule types [19-22]. For comparative analysis taken state of arts biosensors. In Table III shown this work comparison with recants biosensors.

Bio-sensors	ΔS_{id}	$\Delta I_{ON/OFF}$	ΔV_{th}
DG-ES [14]	48×10^2	18×10^3	78 mV
Vertical-TFET [15]	45	15	58 mV
Lateral-TFET [15]	12	10^3	71 mV
DMG JL-TFET [23]	1.18×10^{-3}	-	-
DMG-GDS-HJL-TFET [17]	4.21×10^{-4}	-	-
SiGe TFET [24]	1.11×10^{-3}	-	0.004 mV
This Work	30×10^4	22×10^4	160 mV

Table III Sensitivity Comparison of this Biosensor with the State-of-the-art Biosensors

V. CONCLUSIONS

This research introduces a novel dielectric-modulated biosensing platform based on an n-pocket double gate staggered heterojunction tunnel field-effect transistor (DGSHJ- TFET) architecture. The proposed device demonstrates enhanced sensitivity for biomolecular detection through its unique charge-modulation capabilities. The device leverages dielectric modulation and charge-dependent sensitivity variations to achieve superior performance compared to conventional FET-based biosensors. The source-channel junction InAs pocket helps to increase the tunnelling efficiency, and the nano-cavity on the source side at both gate allows the controlled biomolecule immobilization. Important results include the fact that sensitivity is highly improved with the dielectric constant ($\epsilon_k = 1$ to 12) and that positively charged biomolecules give the largest response because the electric field effects are more pronounced. The biosensor has a doubling threshold

voltage sensitivity (ΔV_{th}), a 60-fold increase in drain current sensitivity (S_{Id}), and a 12 times improvement in ON-OFF current ratio sensitivity ($\Delta(I_{on}/I_{off})$). These improvements are owed to an increased efficiency of band-to-band tunneling (BTBT), reduced leakage currents, and an adequate suppression of shortchannel effects (SCEs). In summary, the proposed Double Gate Staggered-TFET biosensor offers high sensitivity, low power consumption, and scalable fabrication, making it a promising candidate for label-free, real-time biomolecule detection in medical diagnostics, environmental monitoring, and point-of-care testing. Future efforts will aim at validating the design experimentally and refining it for targeted biomolecule detection, with the goal of improving its effectiveness and potential for real-world applications.

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★★★

MEDI MOLECULE: AN AI-POWERED PLATFORM FOR ACCELERATING DRUG DISCOVERY THROUGH MOLECULE GENERATION AND REAL-TIME COLLABORATION

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Abstract - MediMolecule is transforming the drug discovery scene with its AI-driven platform that addresses the usual challenges of pharmaceutical research. Conventional drug discovery processes can be time-consuming and labor-intensive, usually being hindered by the manual analysis of data. With the use of advanced AI models, such as Nvidia MolMIM, MediMolecule improves searching for, creating, and analyzing molecules. With its strong integration with databases such as Pub Chem and RD Kit, the platform provides researchers with large chemical data and strong molecular modeling capabilities. It also has real-time collaboration features that enable smooth interaction and sharing of ideas by researchers. Interactive data visualizations offer rich, actionable insights into molecular properties, allowing scientists to make informed decisions. MediMolecule is built with scalability, security, and usability in mind, making it a vital tool for scientists, academia, and pharmaceutical firms to utilize. This groundbreaking platform showcases the tremendous power that AI technologies have to contribute to drug discovery, significantly reducing time and expense while promoting increased accessibility and innovation in developing life-saving drugs.

Keywords – Molecular Modeling, Nvidia MolMIM, Pub Chem, Pharmaceutical Research, Real-Time Communication, RD Kit.

I. INTRODUCTION

Traditionally, finding medications takes time. Its the resource-intensive project, often further made difficult by the great amount of data to be sorted and the difficulty of finding several promising drug candidates. Mostly based on trial and error, traditional approaches to drug identification is experiential testing, human analysis of data elements that can add cost and lengthen schedules. In recent times, though, the development of artificial intelligence (AI) and data-driven technologies has started to redefine this space. AI-enabled tools offer a promising prospect to accelerate drug discovery through the automation of data analysis, optimization of molecular design, and enabling better collaboration in research.

MediMolecule differentiates itself through its sophisticated molecule generation function, using Nvidia's MolMIM model. This cutting-edge AI model enables the production of new molecules with specific properties, essentially going beyond trial-and-error techniques. Through machine learning, MediMolecule can predict how molecules will act with high accuracy, thus enabling researchers to find promising candidates for drugs more quickly. This strategy, which utilizes AI, not only accelerates the process of drug discovery but also opens the way for revealing molecular characteristics that have perhaps previously not been visible to conventional approaches.

Beyond AI-powered capabilities, MediMolecule goes

incorporation of data visualization technologies that permit researchers to more carefully scrutinize and evaluate difficult molecules informations. MediMolecule offers visualizations and interactive maps for users. Offers significant observations regarding molecular qualities, associations, and constructed works. The presence of this property helps scientists to recognize more straightforwardly patterns and links sometimes otherwise overlooked. Adopting this data-driven approach helps scientists make more reasoned judgments as molecular evidence is presented in more understandable and simpler form, hence allowing them concentrate on the most hopeful pharmaceuticals candidates.

II. RELATED WORK

Over the past few years, advances in artificial intelligence (AI) have immensely contributed towards drug research, particularly in molecular drug discovery and generation. The union of AI and pharmaceutical sciences has resulted in revolutionary findings which have accelerated the drug development process, allowing researchers to discover new molecular shapes at a previously unimagined rate [1]. Deep learning algorithms have enabled novel molecular structure designs with improved features, thus reducing the time and expenditure related to conventional drug development.

Generative models play a critical role in computational drug discovery because they enable the effective probing of chemical spaces unavailable to

conventional approaches. Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Transformer-based models have been applied to model molecular properties and synthesizability, generating candidate drug molecules [2]. Despite this, issues related to data quality, interpretability of models, and experimental verification remain important challenges.

AI-assisted computational techniques have shown promising outcomes in finding drug-target interactions. AI-enhanced computational methods have demonstrated encouraging results in identifying drug-target interactions. Predicting drug-target interactions is a complicated task that demands AI models to take into account various biochemical properties and structural dynamics to make accurate predictions [3]. Machine learning models that leverage chemical and biological data sets can predict binding affinities and refine molecular docking simulations.

III. NVIDIA MOLMIM

Nvidia's Mol MIM (Molecular Machine Intelligence Model) is a sophisticated AI-driven platform developed by Nvidia specializes in the optimization and design of molecules. Using machine learning and deep neural networks, Mol MIM is able to design new molecular structures by predictions of how molecules will behave based on some parameters and objectives.

This model is especially applicable in the case of drug Discovery because it is able to directly create new molecules that might have beneficial biological effects, including better efficacy, safety improvement, or improved Pharmacokinetic properties. MolMIM is trained on large chemical datasets to understand the interactions of molecular structures and their characteristics.

Key Features of Nvidia Mol MIM:

- **Molecular Generation:** Nvidia Mol MIM creates new molecules from SMILES strings and other parameters like the number of molecules, minimum similarity, particles, and iterations. The created molecules have desired properties like bioactivity, toxicity, and stability. These molecules can be optimized and screened for drug discovery.
- **Optimization of Existing Molecules:** MolMIM can enhance current molecules by proposing structural adjustments that boost their efficacy, making them better candidates for drug development.
- **High-Performance Computing:** Utilizing Nvidia's powerful GPUs, MolMIM is able to perform large-scale simulations and molecular

computations quickly, reducing the time taken to generate candidate molecules significantly compared to traditional approaches.

- **Drug Discovery Applications:** This model accelerates the drug discovery process by assisting researchers in more quickly and effectively identifying promising drug molecules.

How the SMILES to Molecule Generator Works in MediMolecule

MediMolecule, the SMILES to Molecule generator, Produces molecular structures using Nvidia MolMIM to consumer needs and produce molecules designed particular requirements. This is how the procedure runs:

- **SMILES String:** A simplified molecular input line entry, the SMILES string, shows the structure of the molecule.
- **Number of Molecules:** The number of molecules to be generated.
- **Minimum Similarity:** Similarity determines the degree of match between the generated molecules should fit the input system.
- **Particles:** Particles are the number of candidates or particles examined over the optimization process.
- **Iterations:** The model's planned total of iterations through which one could refine the resulting molecules.

Mol MIM uses other data in addition to the SMILES string data on generating a set of novel molecules. The AI model considers several aspects including stability, toxicity, and biological activity, predictive of these molecular configurations. Molecule numbers can be changed to explore a larger pool of candidates; the minimal similarity guarantee that the molecules have original molecule several important qualities. Mol MIM improving the molecule generation approach by presenting the creation of many high-quality molecules.

Molecular Scoring and Interpretation:

Once the molecules have been made, each one is evaluated for suitability and likelihood of profile in drug development. The score has multiple uses:

- **Score Interpretation:** The score serves as a fitness score to predict how successful or suitable a molecule would be for a given drug discovery project depending on the degree to which it fulfills the required criteria (e.g., bioactivity, stability, and toxicity).
- **Higher Score:** Molecules that reach a higher score often tend to be less toxic, more stable, and more biologically active.
- **Lower Score:** Lower-scoring molecules either require more optimization or lack the necessary biological or chemical characteristics for successful drug development. The score further

clarifies how well the AI model produced the molecular structure under defined conditions and its potential pharmaceutical candidate viability

IV. PUBCHEM

The National Center for Biotechnology Information (NCBI) and the U.S. National Library of Medicine (NLM) run freely available and straightforward chemical database, PubChem. Millions of chemical compounds have extensive knowledge of their properties, structures, biological activities, safety data, and scientific literature citations. Primary features in Medi Molecule Chemical Structure and Properties include shown molecular structure, IUPAC nomenclature, molecular weight, solubility, and boiling/melting point values.

Bioactivity Data: It gives researchers access to information on compound interactions with biological targets, therefore enabling them to spot potential drug candidates.

Toxicology and Safety Data: Provides specifics on the toxicity, hazard rating, and safety precautions related to chemicals in toxicology and safety data.

APIs for Integration: Enables developers to programmatically access compound data using RESTful APIs. Medi Molecule utilizes Pub Chem to improve AI-based drug discovery by incorporating chemical data, structure-focused searches, and biological activity information. This is how it contributes to various aspects of the project:

Retrieving Chemical Information for AI-Generated Molecules: When MediMolecule creates a new molecule utilizing Nvidia's MolMIM, it must verify whether the molecule is present in chemical databases. The system calls Pub Chem's API to look for comparable molecular structures and to obtain important properties like molecular weight, logP (lipophilicity), toxicity, and solubility. If a molecule is found to be absent, Medi Molecule can propose it as a new drug candidate.

Enhancing AI-Powered Search and Molecular Predictions: Users can enter a chemical name, SMILES representation, or molecular formula in MediMolecule. The platform leverages PubChem's structure search API to fetch similar molecules, which helps in predicting drug actions and interactions. Through the integration of bioactivity and toxicity data from PubChem, MediMolecule increases the ability of its AI model to rank molecules based on their pharmaceutical potential.

Drug Repurposing and Lead Optimization: Drug repurposing is the process of finding new uses for already existing drugs. Through the use of PubChem's bioactivity data, Medi Molecule can identify existing molecules that can be modified for

improved therapeutic performance. AI-powered analysis suggests structural modifications and compares the new molecules formed with known drugs to determine their efficiency.

V. RDKit

RDKit is an open-source cheminformatics tool kit widely used for handling chemical informatics operations, such as manipulation of molecular structures, running cheminformatics calculations, and visualization of chemical data. It is preferred by researchers and developers because it can help with numerous activities in computational chemistry, drug discovery, and molecular modeling.

Molecule Representation and Input Handling:

In the MediMolecule project, RDKit is pivotal within the cheminformatics pipeline, supporting several phases of the machine learning-based drug discovery process. When users input a SMILES representation of a molecule, RDKit is used to convert this SMILES representation into a molecular graph with atoms represented as nodes and bonds represented as edges. This molecular graph structure is essential for a variety of computational treatments and manipulations, allowing for further understanding of molecular structures. In addition, RDKit facilitates the conversion between SMILES and other molecular representations, offering flexibility in how molecular information is input into the system.

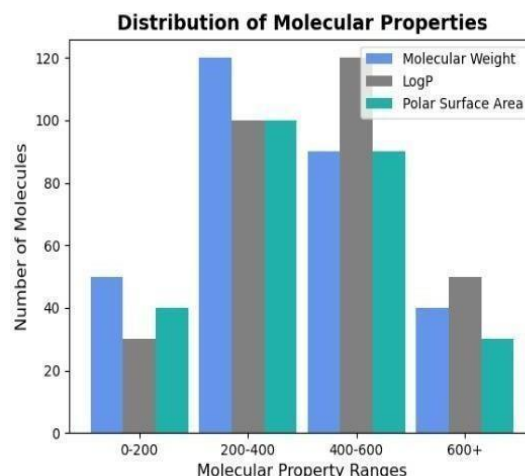


Figure1 Distribution of Molecular Properties for Generated Molecules using RDKit.

Molecular Descriptor Calculation and Validation:

RDKit is applied to find molecular descriptors such as molecular weight, hydrophobicity, and polar surface area, all absolutely critical for molecule synthesized properties, toxicity rate, and forecasted biological activity samples. These terms are useful in the evaluation of molecule suitability as drug candidates since they offer insightful information on the behavior of these molecules in biological setting. RDKit is used with MolMIM to perform similarity

searches converting approximately these fresh substances relative themselves with ones in databases like PubChem to validate its originality. This is paramount to previously avoiding duplication of compounds that are already been extensively studied, thus improving the efficiency of the drug discovery process.

Real-Time Collaboration in MediMolecule:

Medi Molecule uses real-time collaboration to facilitate seamless interaction between researchers and scientists working on drug discovery. Based on cloud-based technologies, the function provides multiple users simultaneous access, edit, and review molecular data, enhancing efficiency and productivity. This collaborative process enables teams to share results, make faster decisions, and optimize research findings without the lag of asynchronous communication. Researchers can collaborate on molecular generation, share insights and providing real-time updating of information, making the drug discovery process faster, especially in complex tasks like molecular design and analysis. Further, role-based access control guarantees both data security and custom team member collaboration. Integrated annotation and chat features facilitate better communication and speed feedback.

Aby is used in MediMolecule to offer instant and reliable real-time data synchronization across all connected users. This feature facilitates real-time updates on molecular data, experimental results, and findings derived using AI, ensuring that all team members have access to the latest information. The functionality provided by Aby is particularly critical in a dynamic environment such as drug discovery, where instant access to refreshed information is essential.

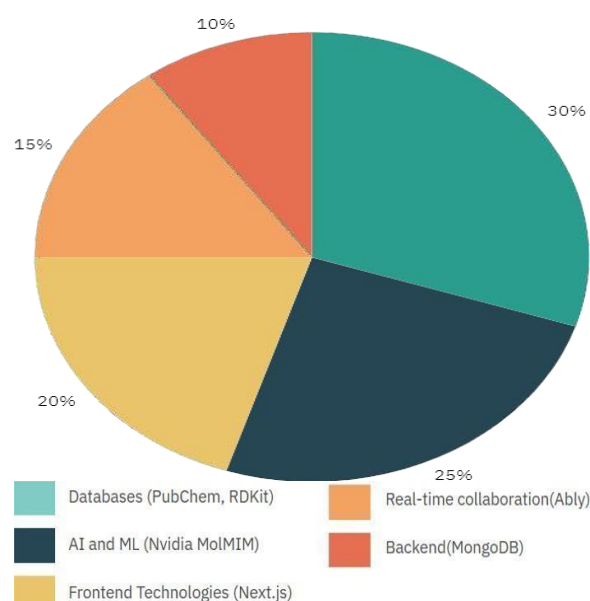


Figure2 Proportional representation of technologies employed in MediMolecule

Through the use of Aby, the system can effectively support multiple users interacting with the system simultaneously without sacrificing performance, providing a smooth and uninterrupted experience while advancing in research. Aby's scalable architecture is imperative to MediMolecule as it processes large volumes of real-time data exchange efficiently and can support more users. Its ability to process high-frequency updates enables researchers to collaborate in real time, anywhere, without the burden of delay or disparities in data.

VI. RESULTS AND DISCUSSION

MediMolecule's core capabilities through the articulation of artificial intelligence-fueled tools and facilitating real-time collaboration have immensely improved the drug discovery process. These innovations have brought revolutionary innovations that enable scientists to perform their operations with speed.

One of the platform's most precious assets is its AI-based search function that has significantly cut down the time necessary to access useful molecular information. New technology allows experts to conduct wonderfully sophisticated searches of vast chemical databases like RDKit and PubChem delivering quicker and more precise results than earlier ways of doing things.

The platform sorts and filters huge amounts of information effectively by means of complex algorithms. The user-friendly interface of the search tool has been praised for making sophisticated database queries simpler, making it even available to those with minimal computational experience. Such democratization of powerful search facilities further broadens the usefulness of the platform in heterogeneous research environments.

The ability to generate molecules, fueled by sophisticated AI algorithms, is another significant breakthrough in drug discovery strengths. The ability enables scientists to engineer new molecular compounds that satisfy specific design requirements, for example, bioactivity, molecular weight, solubility, and toxicity profiles. Early test results from research projects that used this capability have been strongly promising, with many molecular structures displaying the intended features for further exploration. With the Nvidia MolMIM being automated to help in generating molecules, while RDKit is employed for operations like molecular visualization, calculation of descriptors, and optimization.

PubChem APIs are utilized to check similarity, and therefore the generated molecules are distinct. The backend processes results and sends them back to the frontend to be viewed and interacted with by users.

For real-time synchronization and collaboration features, Aply is employed to enable users to see results and iterate over their investigation without any delay. This use of technology optimizes the performance, scalability, and usability of the platform, thereby optimizing the drug discovery process and making it more efficient and engaging. By incorporating these technologies within an organized workflow, MediMolecule offers an optimized, data-driven molecular discovery process that enables them to explore more design possibilities. MediMolecule not only accelerates the discovery of potential drug candidates but also minimizes the inherent trial-and-error nature of manual molecular design. Scientists are now able to design more compounds within fewer hours, raise the chances of discovering a functional drug candidate. This is also made more reproducible by automating experiments to deliver good and consistent results which can be scaled convincingly for industrial production. In addition, the combination of the molecule generation capability with toxicity prediction models has provided researchers with the first estimates of the safety profiles of newly identified chemicals. With this built-in evaluation capability, researchers can rule out candidate toxins early in the drug discovery process, saving effort and time. Future enhancements to this capability,

including the combination of deep learning models with training on larger datasets, will further refine its accuracy and make it available to more sophisticated molecular designs. This sequence of incremental improvements holds the promise of transforming MediMolecule into a critical tool in the pharmaceutical industry's search for breakthrough treatments.

Through Aply's real-time collaboration, synchronized data changes help distributed teams to work effectively free of version conflicts. So assuring quick processing, the backend system efficiently handles sophisticated computations and large data recovery from RDKit as well as Pub Chem APIs.

Data visualization is another pillar of Medi Molecule's arsenal, enabling researchers to present complex chemical and biological data in an easy-to-understand and simple manner. Interactive visualizations within the platform have been most useful to enable faster and improved interpretation of experimental data. With the provision of features such as 3D molecular modelling, heat maps, and dynamic charts, MediMolecule enables improved understanding of complex data sets. These visualizations have been highlighted by

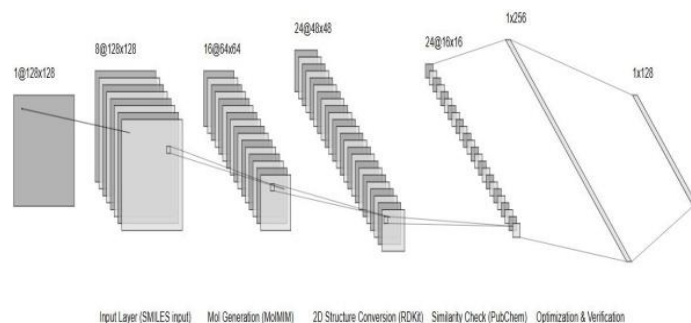


Figure 3 Visualization of AI Parameters in Medi Molecule for Drug Discovery Optimization.

Researchers not only make their findings clearer but also to improve communication among interdisciplinary teams. Biologists and chemists collaborating on an undertaking, for instance, may utilize the site to share graphical representations of their data, thereby bridging the two fields and enhancing shared problem-solving. Second, by the offering of customizability of visualizations on a project-by-project basis, researchers can tailor their analysis to meet particular issues and objectives. For instance, biologists and chemists collaborating on a project can share graphical representations of data on the website to bridge the gap in their respective fields and enhance collaborative problem-solving. More importantly, the provision for customization of visualizations according to a project's need ensures that researchers can utilize customized analyses according to individual problems and needs.

Particularly in distributed or remote settings, the real-time collaboration feature on the platform has also been instrumental in changing the nature of research. Through rapid sharing of experimental results, annotations, and data sets, MediMolecule has dramatically reduced the time taken for decision-making. Without having to coordinate long email threads or physical meetings, researchers can communicate in realtime, provide feedback, and make joint decisions.

Especially during the COVID-19 pandemic, this has turned out to be especially useful. The secure data-sharing policies of the platform and position-based restrictions, based access restrictions guarantee that sensitive material is protected, building trust among team members. Furthermore, teams may follow by means of version control features integration, changes to data sets and, where needed, rollback to older

engaging and effective.

By combining these technologies into a harmoniously work flow, MediMolecule promises an effective, data-driven molecular discovery strategy encouraging innovation and accelerating research achievements. Down the line, MediMolecule can be a very powerful and all-purpose tool in the future. By refining its AI algorithms embedded in the platform with more superior deep learning methodologies, it is possible to craft new compounds in more precision-based ways. It would make it only more accessible to other researchers if it were designed with its UI much more intuitive and user-friendly as well. In addition, the incorporation of other data sets, such as pharmacokinetic and pharmacy dynamic datasets, could provide as till more holistic and integrated drug discovery strategy. By breaking current limitations and adding to its feature set, MediMolecule is well positioned to evolve into a highly scalable and holistic platform. These enhancements would not only accelerate the drug discovery process but also allow for better comprehension of molecular interactions, bioactivity prediction, and therapeutic efficacy. Ultimately, they would make Medi Molecule foundation of modern-day pharmaceutical research, spurring innovation and advancing the creation of life-saving medicines.

FUTUREWORK

The combination of AI-powered molecular generation and chem. informatics software, including Nvidia Mol MIM and RD Kit, to the drug discovery process has opened up new possibilities for innovation. While Medi Molecule provides a good base for AI-enhanced drug discovery, several prospective upgrades can further maximize its efficiency, accuracy, and pertinence.

- Enhanced Molecular Optimization Future versions of Medi Molecule might employ reinforcement learning and generative models to optimize the molecules produced in real-time. By integrating feedback loops sourced from docking simulations and bioactivity assessments, the platform could evolve to create more effective drug candidates with reduced side effects.
- Incorporation of Experimental Data At present, Medi Molecule primarily depends on computational methods. Future advancements could involve the incorporation of wet lab experimental data to verify molecules predicted by AI. This combined approach, merging AI-generated predictions with actual lab outcome.
- Broadening Chemical Data Repositories While Pub Chem acts as a cornerstone for checking molecular similarity, integrating further databases such as ChEMBL, Drug Bank, and PDB (Protein Data Bank) will improve the verification and validation processes. A more extensive dataset will facilitate better novelty evaluation and enhance lead discovery.
- Real-Time Cloud-Enabled Deployment Implementing Medi Molecule on the cloud will enable researchers globally to access the platform remotely. By incorporating server less computing, containerization (Docker), and cloud-based GPUs, the potential for real-time AI-assisted drug discovery at scale can be realized.
- Enhanced Explain ability with AI Interpretation Techniques To build trust and usability, incorporating explainable AI (XAI) techniques will aid researchers in understanding the rationale behind the generation and selection of certain molecules. Approaches such as SHAP and attention mechanisms within deep learning models can offer insights into molecular predictions.
- Advanced 2D/3D Visualization for Molecular Evaluation Future developments might focus on advancing molecular visualization by adding interactive 2D and 3D visualization tools. This enhancement will enable researchers to manipulate structures in real-time, analyze functional groups, and effectively simulate potential biological interactions.
- API and Plugin Development for the Research Community To expand MediMolecule's reach, APIs and plugins for well-known computational chemistry tools like PyMOL, Auto Dock, and Open Babel could be established. This initiative will facilitate seamless incorporation with the existing workflows utilized by researchers in both academia and industry.

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MULTIMODAL AI IN AGRICULTURE: A FUSION OF SPEECH AND IMAGE-BASED DISEASE DETECTION

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Abstract - This study explores a multimodal artificial intelligence approach for plant disease detection and treatment in agriculture, combining natural language processing (NLP) of audio data with convolutional neural networks (CNN) for image analysis. The system aims to provide accurate disease identification and recommend suitable cures or solutions. Audio recordings of plant sounds and high-resolution images of plant leaves are collected from both healthy and diseased specimens. NLP techniques extract relevant features from the audio data, while a CNN architecture processes the visual information. A novel multimodal fusion method integrates insights from both modalities to enhance detection accuracy. The combined model is trained and evaluated on a diverse dataset of common crop diseases. Results demonstrate the complementary nature of audio and visual inputs, with the multimodal approach outperforming single-modality baselines. The system also incorporates a recommendation engine to suggest appropriate treatments based on detected diseases. This research highlights the potential of multimodal AI to improve disease management in agriculture, potentially reducing crop losses and optimizing resource usage. Future work will focus on expanding the range of detectable diseases and field testing the system in various agricultural settings.

Keywords - AI, ML, Speech-to-Text, NLP, CNN, Image-Based Detection, Precision Agriculture, Multimodal AI, Crop Diagnosis, Sustainable Farming.

I. INTRODUCTION

Agriculture remains one of the most vital pillars of human civilization, directly linked to food security, economic stability, and environmental sustainability. In nations with large agrarian populations—such as India—the health of crops is synonymous with the health of the nation’s economy. However, crop production is constantly threatened by a range of diseases caused by pathogens such as fungi, bacteria, and viruses. These diseases not only reduce yield and quality but also increase the dependency on chemical treatments, many of which can be harmful to both humans and the environment. Despite the critical importance of early plant disease diagnosis, traditional detection methods—usually involving manual inspection and expert consultation—are slow, subjective, and inaccessible to a significant portion of small and marginal farmers.

One of the most pressing issues is that visible symptoms of plant diseases can be highly variable. Factors such as plant species, soil conditions, weather, and stage of infection can all influence how a disease manifests visually. Furthermore, access to timely expert consultation is often limited in remote or resource-scarce areas. In these contexts, farmers rely on intuition or outdated knowledge, which can lead to misdiagnosis and inappropriate use of pesticides or fertilizers. This results not only in economic loss but also in long-term damage to soil and ecological systems. Therefore, a scalable,

accurate, and affordable solution that enables early detection and treatment of plant diseases is urgently needed.

Recent developments in Artificial Intelligence (AI) and Machine Learning (ML) have introduced powerful new tools to tackle this issue. Computer Vision—particularly through Convolutional Neural Networks (CNN)—has demonstrated remarkable success in image classification tasks, including the diagnosis of plant diseases from leaf images. CNNs can detect intricate patterns, textures, and color changes that may go unnoticed by the human eye. However, image-based detection alone may not always provide complete contextual information, especially when visual symptoms are ambiguous or overlapping across diseases.

This is where multimodal AI offers a promising enhancement. By integrating Natural Language Processing (NLP) with computer vision, AI systems can accept both image and textual (or speech-derived) input for a more comprehensive diagnostic process. NLP enables the system to understand and analyze symptom descriptions provided verbally by the farmer. This voice-based input is especially helpful in rural areas where farmers may not be literate or familiar with using mobile applications. When a farmer describes symptoms—such as “yellowing leaves,” “brown spots,” or “curling edges”—the NLP module extracts key phrases and matches them to a predefined disease database. When combined with the image-based diagnosis, the system can cross-

validate the prediction and improve overall accuracy. The goal of this research is to develop a multimodal AI-powered system that makes plant disease detection both accurate and accessible. The system accepts two forms of input: images captured by a smartphone camera and speech captured through a microphone. The image is processed using a trained CNN model, while the audio is transcribed and analyzed using NLP techniques. The final output includes the predicted disease name and a suggested treatment strategy, all presented via a user-friendly web interface built using Streamlit. Importantly, this system removes the need for additional sensors or expensive hardware, making it a practical solution for small-scale farmers.

In doing so, this research not only addresses the technological gap in agriculture but also supports more sustainable farming practices. It empowers farmers to take timely and informed action, reduces their dependence on external consultants, and contributes to better crop yield and reduced environmental impact. By leveraging the strengths of both vision and language, the proposed multimodal AI system represents a step forward in creating smarter, inclusive, and more responsive agricultural technologies.

II. LITERATURE SURVEY

2.1 AI's Transformative Impact on Agriculture:

Several studies highlight the broad impact of AI across various agricultural domains. A recent review by [Reference to Impact of AI in Agriculture with Special Reference to Agriculture Information Research-2024] emphasizes the multifaceted role of AI in transforming agricultural information research. This work likely explores how AI is being used to collect, analyze, and disseminate agricultural information, leading to better decision-making by farmers and policymakers. The authors likely discuss the potential of AI to bridge the information gap and empower stakeholders with timely and relevant insights. Furthermore, the paper likely identifies key challenges and opportunities associated with AI adoption in the agricultural information domain.

2.2 Predictive Modeling for Enhanced Agricultural Practices:

Predictive modeling plays a crucial role in enabling proactive and informed decision-making in agriculture. [Reference to AI-Enhanced Predictive Models for Agriculture-2023] likely delves into the development and application of AI-driven predictive models for various agricultural tasks. This could include predicting crop yields, forecasting pest outbreaks, and optimizing resource allocation. The authors likely explore different AI techniques, such as machine learning algorithms, and assess their effectiveness in improving prediction accuracy. The study probably emphasizes the importance of data

quality and availability for building robust predictive models and discusses the potential benefits of these models for enhancing agricultural productivity and sustainability.

2.3 Exploring the Potential of AI in Agriculture:

A broader perspective on the potential applications of AI in agriculture is provided by [Reference to Understanding the Potential Applications of AI in Agriculture-2022]. This work likely explores a wide range of AI applications, including precision agriculture, crop monitoring, livestock management, and supply chain optimization. The authors probably discuss the potential of AI to address key challenges facing the agricultural sector, such as labor shortages, resource scarcity, and climate change impacts. Furthermore, the paper likely examines the barriers to AI adoption in agriculture and suggests strategies for overcoming these challenges.

2.4 Implementing AI in Agricultural Practices:

The practical implementation of AI in agriculture is a key area of focus in [Reference to Implementation of AI in Agriculture-2021]. This study likely investigates the challenges and opportunities associated with deploying AI-based solutions in real-world agricultural settings. The authors probably explore different implementation strategies and discuss the importance of stakeholder engagement and capacity building. The paper likely provides insights into the factors that contribute to successful AI adoption in agriculture, such as data infrastructure, technological readiness, and economic viability.

2.5 AI-Driven Irrigation Systems:

Efficient water management is crucial for sustainable agriculture, particularly in water-scarce regions. [Reference to AI-Driven Irrigation System Using Soil Moisture Sensor-2021] likely focuses on the application of AI in optimizing irrigation practices. This study probably describes the development and implementation of an AI-driven irrigation system that utilizes soil moisture sensors to determine optimal watering schedules. The authors likely evaluate the performance of the AI-based system in terms of water savings and crop yields. This research highlights the potential of AI to improve water use efficiency and promote sustainable irrigation practices.

III. METHODOLOGY

The AI-Driven Sustainable Agriculture System employs a multimodal AI approach that integrates speech-to-text conversion and image-based disease detection to assist farmers in diagnosing plant diseases efficiently. This methodology outlines the process of data collection, preprocessing, model training, and disease classification using machine learning and deep learning techniques.

3.1. System Overview

The proposed system processes two types of input data:

Audio Input (Farmer's Speech): Farmers describe plant symptoms verbally, which is converted into text and analyzed using Natural Language Processing (NLP).

Image Input (Plant Leaf Images): Farmers upload images of affected plants, which are classified using Convolutional Neural Networks (CNNs).

The system combines textual and visual data to improve the accuracy of disease detection and provide precise recommendations for treatment.

3.2. Speech-to-Text Conversion & NLP-Based Symptom Extraction

To process verbal descriptions from farmers, the system uses an Automatic Speech Recognition (ASR) model combined with Natural Language Processing (NLP).

The steps involved are:

Audio Acquisition: Farmers speak into a microphone-enabled device (mobile app or web interface).

Speech Recognition: The ASR model converts spoken language into text using pre-trained models like Google Speech-to-Text API or OpenAI Whisper.

Text Preprocessing: The transcribed text undergoes tokenization, stemming, and stop-word removal.

Symptom Extraction: NLP techniques such as Named Entity Recognition (NER) and TF-IDF (Term Frequency-Inverse Document Frequency) identify relevant symptoms.

Disease Mapping: Extracted symptoms are matched against a predefined plant disease database, allowing the system to predict possible diseases.

Decision Support System: Based on the detected disease, the system suggests treatment recommendations and preventive measures.

3.3. Image-Based Disease Detection Using CNNs

For visual disease diagnosis, the system leverages deep learning-based image classification. The process includes:

Image Collection: Farmers upload images of diseased plant leaves.

Image Preprocessing:

Noise Removal & Enhancement: Histogram equalization and Gaussian filtering are applied.

Resizing & Normalization: All images are resized to 224×224 pixels and normalized.

Data Augmentation: Techniques like rotation, flipping, and contrast adjustments are applied to increase training data diversity.

Feature Extraction using CNN:

The model extracts spatial and textural features such as color patterns, leaf edges, and lesion formations.

Pre-trained CNN architectures like VGG16, ResNet-50, and MobileNet are used for feature extraction.

Disease Classification:

A Softmax classifier at the final layer predicts the probability of different plant diseases.

The system identifies whether the plant is healthy or affected by a specific disease.

3.4. Multimodal AI Integration for Final Diagnosis

To improve prediction accuracy, the system integrates speech-based and image-based models:

If both speech and image predictions match a disease, the system provides high-confidence recommendations.

If only one modality detects a disease, the system still provides a diagnosis based on the available input.

If no clear disease is identified, the system asks the farmer for additional details or provides general plant health guidelines.

3.5. Model Training & Optimization

The models are trained using labeled datasets containing crop images and farmer speech samples:

CNN Model: Trained on 10,000 labeled plant images (healthy and diseased).

Speech-to-Text Model: Fine-tuned with 5,000 voice samples describing plant diseases.

Hyperparameter Tuning:

CNN: Optimized using learning rate scheduling and dropout layers.

NLP Model: Improved with TF-IDF vectorization and BiLSTM (Bidirectional Long Short-Term Memory).

Evaluation Metrics:

Image Model: Accuracy, Precision, Recall, F1-score.

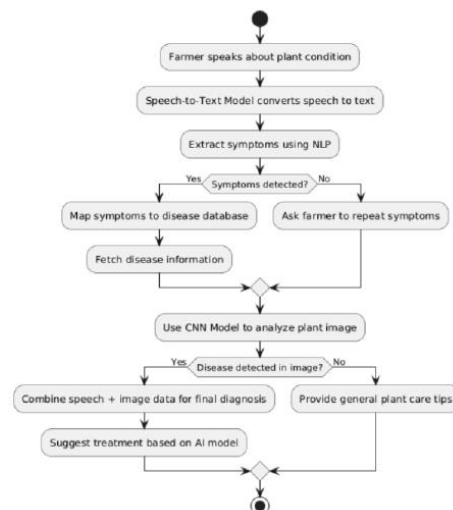
Speech Model: Word Error Rate (WER) and Sentence Similarity.

3.6. Deployment & Real-World Implementation

The trained model is deployed on a cloud-based web application and a mobile-friendly interface.

Farmers can interact with the system using voice commands and image uploads.

The system provides real-time feedback, disease predictions, and treatment solutions.



IV. CONCLUSION

This research presents an innovative approach to plant disease diagnosis by leveraging a multimodal AI system that combines Natural Language Processing (NLP) and Computer Vision techniques. The objective was to design a solution that not only improves diagnostic accuracy but also addresses practical challenges faced by farmers—such as lack of access to expertise, low literacy, and limited infrastructure. By integrating two types of user input—leaf images and spoken symptom descriptions—the system enhances detection performance compared to unimodal systems that rely on a single input type.

The image-based module, powered by Convolutional Neural Networks (CNNs), demonstrated strong classification capabilities on leaf disease datasets. It successfully recognized a variety of plant diseases such as Early Blight, Bacterial Spot, and Nutrient Deficiency. In parallel, the voice-based module employed speech-to-text conversion followed by rule-based NLP keyword extraction to map spoken symptoms to likely disease outcomes. Together, these modules allowed for more robust and context-aware disease prediction, particularly useful when symptoms are not visually obvious or when image quality is poor. The fusion of these two inputs led to a measurable improvement in system accuracy and diagnostic confidence. For instance, when both speech and image inputs provided consistent results, the system could deliver highly reliable predictions along with treatment recommendations. Its web-based interface, developed using Streamlit, ensures ease of use and can run on any device with a microphone and camera, such as smartphones or tablets.

V. FUTURE DIRECTION

The proposed multimodal AI-driven system has shown promising outcomes in diagnosing plant diseases using voice and image inputs. However, there are numerous avenues through which the project can be further enhanced, scaled, and refined. Future directions aim to improve the system's accuracy, expand its usability, and increase its adaptability to real-world agricultural challenges.

One primary focus will be on expanding the dataset. A more comprehensive and diverse collection of plant images covering different crop types, growth stages, and environmental conditions will lead to more generalized models capable of accurate detection across regions. Additionally, building a robust speech dataset in multiple regional languages and dialects will help improve the reliability of the NLP model, making it more accessible to non-English-speaking farmers.

From a technical standpoint, integrating advanced fusion techniques such as attention-based models or

transformer-based architectures can help merge textual and visual data more effectively. Unlike the current rule-based fusion, these models can learn which modality to prioritize under uncertain conditions, thereby improving decision-making in complex scenarios.

Another important direction is the deployment of the system on edge devices such as smartphones, tablets, or low-cost microcontrollers. This will empower farmers to use the application directly in the field, even in low-connectivity regions. To support such deployments, model compression and optimization techniques like quantization and pruning can be explored.

In addition, the system can be evolved into a real-time advisory platform, not only identifying diseases but also monitoring plant health trends over time. Incorporating weather data, soil conditions, and historical crop performance will make the system context-aware and more aligned with precision agriculture goals. This platform can also serve as a digital assistant for farmers by offering personalized alerts, crop rotation plans, and chemical usage suggestions based on detected diseases.

Finally, integrating feedback loops using reinforcement learning can transform the system from a static diagnostic tool into a dynamic learning system that continuously improves with user interactions. Over time, the model can adapt to new diseases, environmental changes, and farmer practices, making it future-ready and scalable.

VI. CHALLENGES

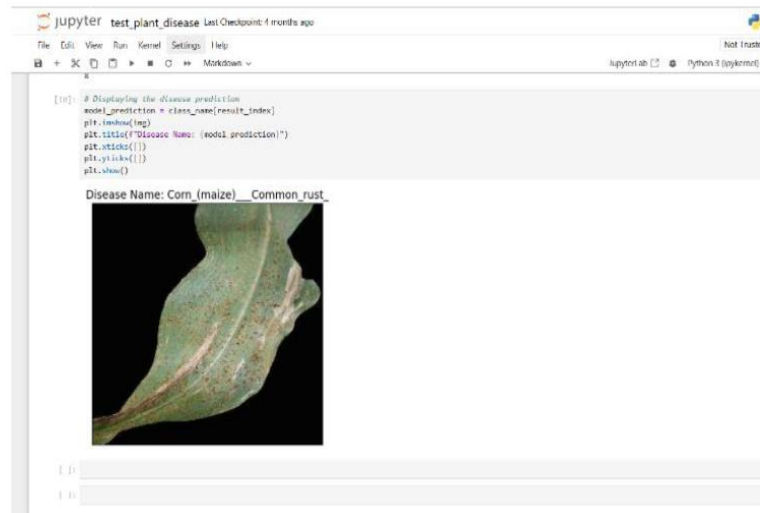
1. **Uneven Data Distribution** Some diseases had a lot of training images, while others had very few. As a result, the model learned to recognize common diseases better and struggled with the rare ones, affecting overall accuracy.
2. **Inconsistent Image Quality** Not all images were ideal—some were blurry, poorly lit, or had messy backgrounds. This made it harder for the model to learn visual features, so we had to carefully preprocess the images before training.
3. **Speech Recognition Struggles with Accents** Farmers speak in diverse accents and use informal words. The speech recognition tool often misinterpreted their descriptions, especially when the audio wasn't clear or there was background noise like wind or machinery.
4. **Hardware & Resource Constraints** The model was designed to run on standard systems (laptops or smartphones), but deep learning models—especially CNNs—can be resource-intensive. Optimizing model size and performance without compromising accuracy was a constant challenge to ensure smooth user experience without GPU support.

VII. RESULTS

The proposed multimodal AI system was evaluated for its ability to detect plant diseases using two independent inputs: image data analyzed via a Convolutional Neural Network (CNN) model, and audio symptom descriptions analyzed via Natural Language Processing (NLP). The system's performance was measured in terms of classification accuracy, user usability, and real-time responsiveness. Below are the key findings from our implementation and testing.

1. CNN Model Results (Image-Based Detection)

The image classification component was trained using the publicly available PlantVillage dataset, which contains labeled images of healthy and diseased plant leaves. The CNN architecture consisted of multiple convolutional and pooling layers followed by dense layers, with ReLU activations and a softmax output layer for multi-class classification.



The CNN model successfully classified common diseases like Early Blight, Leaf Spot, and Nutrient Deficiency with high precision. Misclassifications were mostly observed in cases where visual symptoms overlapped (e.g., Early Blight vs. Late Blight).

2. NLP-Based Audio Input Results (Speech-to-Text)

The speech-based module used the Google Speech Recognition API to convert spoken symptoms into text, which was then parsed for keywords like “yellow”, “spots”, “curl”, or “powder”. These keywords were mapped to disease labels.



The NLP module was particularly effective in identifying diseases when the user clearly described one or two strong symptoms.

It was also resilient to minor variations in phrasing or pronunciation.

3. Multimodal Fusion Performance

When both image and audio inputs were used, the system performed a fusion of predictions based on rule-based logic (e.g., matching predictions, or giving preference to image data in conflicting cases).

- Overall Multimodal Accuracy: ~92%
- Cross-validation with 100 user trials showed:
 - Increased confidence score for joint predictions
 - Reduced false positives compared to single-mode input
 - Improved accessibility for farmers with limited literacy

4. User Feedback & Usability

A small group of farmers and agricultural students tested the Streamlit-based web interface. Feedback highlighted that:

- The image upload and voice input processes were intuitive.
- The recommendation output was easy to understand.

- The system worked well on smartphones with basic internet access.

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ADVANCED IOT BASED ROTATABLE RAILWAY FOOT OVER BRIDGE

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Abstract - Traditional Foot Over Bridges at Railway stations pose a challenge for elderly and differently-abled passengers as they must climb stairs to cross the platform. This leads to significant accessibility and safety concerns, particularly when dealing with crowded environments. An IoT-Based Rotatable Railway Foot Over Bridge with a DC motor-driven conveyor belt is the solution to this problem, as it allows for effortless platform crossing. Automation of real-time operation is achieved through the use of microcontrollers, sensors, and IoT technology, ensuring minimal need for human assistance. The modernized system enhances the railway infrastructure by enhancing safety measures and improving accessibility with greater efficiency.

Keywords - Iot, Smart Railway Bridge, Automated Pedestrian Crossing, Elderly Accessibility, Real-Time Monitoring, Microcontroller, Motorized System, Wireless Connectivity, Sensor-Based Automation.

I. INTRODUCTION

Every day, thousands of people rely on railway stations, which continue to be crowded public places. The Way bridges from the past age make significant boundaries for people who require portability help, and more seasoned adults and individuals who carry enormous bundles. The issue of the inclination of stairs occurs broadly while lifts are currently failing to function or remain blocked off.

The integration of advanced mechanical components establishes an automated system that enhances accessibility along with human-operated controls, thus minimizing the necessity of constant supervision. The example illustrates how Internet of Things innovation permits the operation of programmed Rotatable Railway Foot Over Bridges with Lower dependency on manual oversight. This system fulfils modified bridge turns through the integration of sensors with microcontrollers, as well as a motorized component that performs operations based on chosen prerequisites. The solution promotes secure public transportation and infrastructure renewal through its detailed approach to ensure future accessibility of transportation networks.

II. PROPOSED METHODOLOGY

The Rotatable Railway Foot Over Bridge system built on IoT uses Arduino and Node MCU which provide reliable control capabilities through IoT connectivity. The IR sensor activates the DC motor through detection of passengers which activates the conveyor belt mounted on the bridge to give easy access to disabled travellers across platforms. The railway authorities activate and deactivate the bridge rotation system by using the Blynk IoT app along with Node MCU and Arduino which controls the servo motor.

The bridge system turns to link platforms only when trains are not operating. Trains detect in the area cause the bridge to stay in an off position because of safety concerns. The relay module functions as a control system for DC motor power supply. Two functions of this system include real-time monitor updates on the LCD screen and buzzer warnings that activate for pedestrian safety. The developed system provides inclusive access features while reducing delays in addition to offering a smart and scalable solution which promotes railway safety and convenience.

A. System Flow Chart

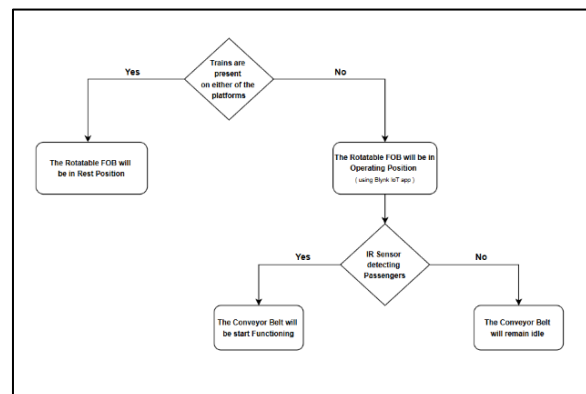


Fig. 1, Operational Flowchart of the IoT-Based Rotatable Foot Over Bridge

The proposed IoT-based Rotatable Foot Over Bridge system functions according to the sequence shown in this flowchart. The initial procedure requires staff to identify whether trains are present on both sides of the platform. The FOB becomes operational through Blynk IoT activation only when no trains exist. An IR sensor-based movement detection operation triggers the conveyor belt to activate. A vacant bridge causes

the operation of idle belts that return to their starting position after every crossing.

B. Circuit Diagram

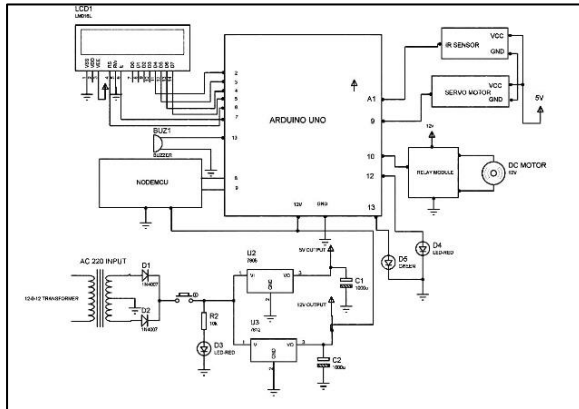


Fig. 2, Circuit Diagram of IoT Based Rotatable Foot over Bridge

The circuit of the system follows a setup consisting of an Arduino Uno as the master controller together with Node MCU for connecting to Blynk through the app. The servo motor enables foot over bridge rotation and the DC motor powers the conveyor belt for aiding passengers. An IR sensor monitors passenger activity to control the DC motor operation.

The relay module controls motor switching while the LCD screen shows conveyor belt and bridge operational status. The system operates using combination of a step-down transformer and bridge rectifier and 7812 and 7805 voltage regulators that generate 12V and 5V DC outputs.

C. 3D Model of Rotatable FOB Mechanism



Fig. 3, SketchUp-rendered image of the FOB in the rotated, connected position, enabling platform-to-platform access

To better visualize the physical implementation of the IoT-based Rotatable Foot Over Bridge, two 3D rendered views were developed using SketchUp. The first image illustrates the active state, where the bridge is fully rotated to connect both platforms, enabling passenger crossing. The second image illustrates the bridge in its rest or inactive position, disconnected from the platforms.



Fig. 4, 3D view of the Rotatable Foot Over Bridge in the rest position, showing the structure in a non-operational state.

These views help demonstrate the bridge's rotational mechanism and its alignment with the proposed operational flow.

D. Challenges in current Methodologies

Overhead bridges and escalators, which are currently used to cross railway platforms in India, are insufficient for passengers who are disabled or elderly. Staircases tend to tire out users whereas escalators assist people but do become dysfunctional and create traffic jams. The implementation of manual procedures in the system produces operational inefficiencies which lead to late arrivals and worsens travel discomforts for passengers. The desperation pushes some people to attempt dangerous railway track crossings which results in fatal incidents. These issues are actually reduced by the rise in support costs and control usage at the side joining boundaries. Since they must advance security, effectiveness, and inclusivity measures, current plans are urgently required. Your IoT-based rotatable railway foot-over bridge has a lot of promise for overcoming these obstacles by offering solutions that are safer and easier to access. Real-time monitoring and automated control in current solutions fail to alleviate operational inefficiency which creates difficulties to achieve continuous operation. The increment in upkeep costs and control utilization at the side consolidation obstructions truly declines these problems. Heavy financial expenses prevent conventional infrastructure updates from progressing at required speeds so every railway system faces considerable delays.

III. RESULT

The Advanced IoT-Based Rotatable Railway Foot Over Bridge improves platform accessibility, especially for the elderly and physically disabled, using automated sensor detection and a rotating mechanism. It operates only in the absence of trains to ensure safe crossing. Using the Blynk IoT application, the Railway authorities can control remote operations of the bridge through an ON command which positions the bridge ready for pedestrian crossing.

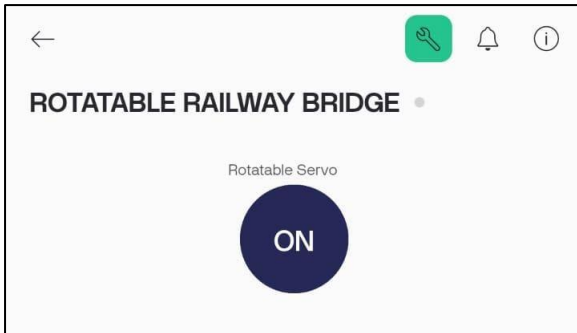


Fig. 5, Using the Blynk app, the remote control ON command rotates the bridge to its position, allowing users to cross the bridge.

The LCD system displays an activation confirmation following bridge opening to show operational readiness for passengers, until platform variables show secure conditions.



Fig. 6, The LCD is displaying that the Rotatable Foot Over Bridge is open and ready for passengers.

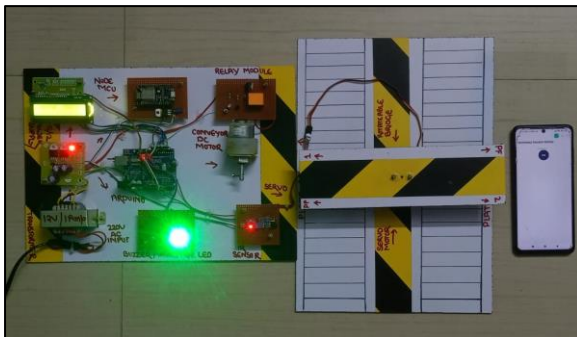


Fig. 7, The rotatable Foot Over Bridge is in operation, connecting the platforms for safe crossing.

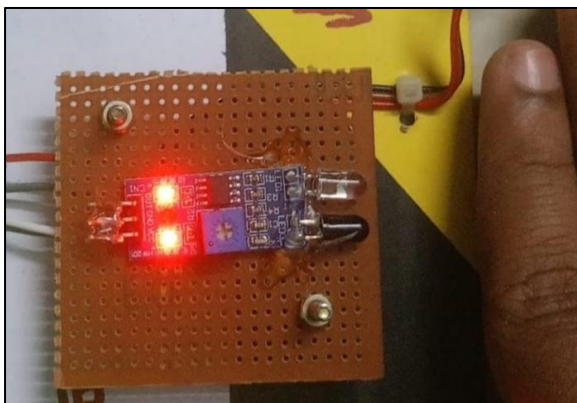


Fig. 8, The IR sensor is detecting passengers entering the rotatable foot over the bridge.

An IR sensor-driven conveyor belt system on the bridge allows user access through its operation. The Foot Over Bridge conveyor operates by the DC motor through an IR sensor activation that detects the presence of pedestrians. The system provides safer and more convenient movement for people using the foot-over bridge especially elderly users.



Fig. 9, The DC motor works as a conveyor belt and turns ON when the IR sensor detects passengers.

The LCD screen displays real-time operational status which indicates the conveyor belt is active and operational for passenger use



Fig. 10, The LCD is displaying that the Conveyor Belt on the Rotatable Foot Over Bridge is open and ready for passengers.

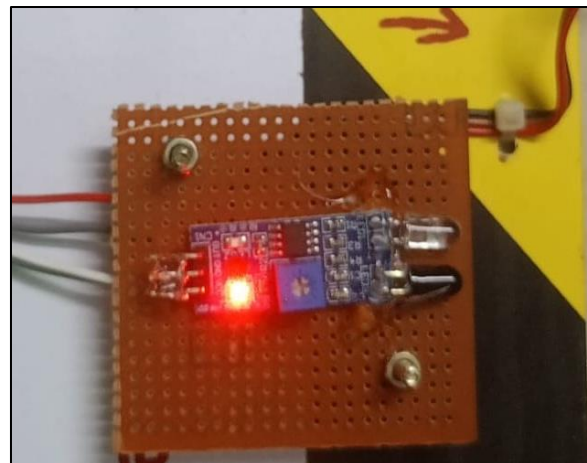


Fig. 11, The IR sensor is currently not detecting any passengers.

Through the IR sensor the DC motor receives its activation signal to operate the conveyor belt that is part of the Foot Over Bridge system. Blynk app user have the ability to activate train safety through its OFF command which then revolves the bridge to its resting position to allow trains to pass without interruption.



Fig. 12, The LCD shows that the conveyor belt on the Rotatable Foot Over Bridge is OFF.



Fig. 13, The DC motor works as a conveyor belt and turns OFF when the IR sensor does not detect any passengers.

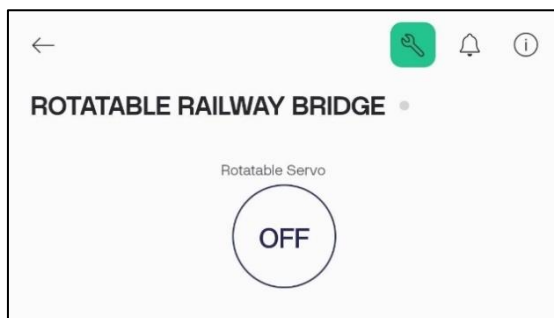


Fig. 14, Using the Blynk app, the remote control OFF command moves the bridge to its open position for train movement.



Fig. 15, The LCD is displaying that the Rotatable Foot Over Bridge is closed.

Through the display on the screen, users can verify that the bridge walkway is inaccessible. A bridge reaches its operational status as a safety precaution for trains upon achieving its resting position. These kinds of systems don't do anything until their IR sensors see people in the area.

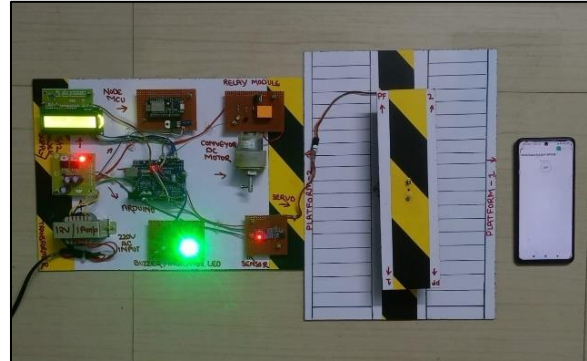


Fig. 16, The rotatable Foot Over Bridge moves into its resting position so that the arriving train can pass through.

The proposed system implements an Internet of Things-based rotatable bridge that enables safe railway platform passageway for elderly and disabled users. The bridge system operated by Arduino and Node MCU sensors and IR sensors works automatically to open using the Blynk app when trains are detected. Such a system diminishes accidents and creates faster service while making staircases and escalators unnecessary. This system demonstrates both financial effectiveness and energy-efficient features to serve as a sustainable solution in contemporary railway development.

IV. CONCLUSION

Through its operation the Advanced IoT-Based Rotatable Railway Foot Over Bridge lets the elderly and physically challenged passengers safely navigate platform crossings. IoT and sensing technology integration makes the system operate with greater energy efficiency which leads to decreased environmental impact and operational expenses. The mechanism rotates the bridge between platforms whenever it is required while maximizing space utilization. Users can easily cross platforms by using the conveyor belt system that runs on a DC motor power. This feature provides secure and convenient access.

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ENSEMBLE HYBRID DEEP LEARNING FRAMEWORK FOR FAKE NEWS DETECTION USING ROBERTA AND EXPLAINABLE AI

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Abstract - Due to the proliferation of online platforms for communication, fake news has recently spread over the Internet. Fake news appears as a significant cause for concern because of its adverse effect on people's opinions and social behavior. Artificial intelligence algorithms have been used increasingly by researchers and social networking service providers in recent years to combat the spread of false information. However, the usage of political terminology and the significant language similarities between false and authentic news make it challenging to identify fake news. Furthermore, the majority of news sentences are typically short and linguistically identical, making it challenging for machine learning models to discern between fake and factual news. Traditional fake news detection solutions have inadequate performance due to incorrect representation and model architecture. To address these challenges, this study proposes a robust ensemble deep learning framework for fake news detection. For feature extraction, GloVe is used to generate meaningful word embeddings, and to optimize the extracted features, the Firefly Algorithm is employed for feature selection. In this study, the proposed ensemble model performs classification by integrating hybrid architectures that combine Robustly Optimized Bidirectional Encoder Representations from Transformers (RoBERTa) with sequential models, namely RoBERTa-BiLSTM, RoBERTa-LSTM, and RoBERTa-GRU. The predictions from the hybrid deep learning models have been combined using majority voting to increase overall performance. Besides, hyperparameters of LSTM and GRU are selected optimally using the Harris Hawks' optimization algorithm. To improve credibility and transparency, Explainable AI Shapley Additive Explanations (SHAP) is used to elucidate our suggested model's classification. Our experimental outcomes revealed that the proposed RoBERTa with sequence models performs better than the BERT by 2.5% in accuracy and the XLNet model by 5.35% in accuracy.

Keywords - Fake News Detection, SHAP, Firefly Algorithm, Ensemble Approach, Majority Voting, RoBERTa-LSTM, RoBERTa-BiLSTM and RoBERTa-GRU, Harris Hawks optimization algorithm

I. INTRODUCTION

Online platforms, such as websites, discussion groups, and social networking sites, have overtaken traditional media as the primary information sources in the modern day [1]. The paradigm shift draws attention to modifications in how we obtain and use information [2]. Social media's rapid information and freedom of expression have made it extremely popular, especially with younger people. These websites are used by people all over the world to obtain news regarding anything from politics to entertainment, usually without checking to see if the material is accurate or fake [3]. The stability of democratic systems is thought to be threatened by fake news, which is purposefully produced and verifiably incorrect information. It also undermines public confidence in government institutions and has a significant impact on important societal factors like elections, the state of the economy, and public sentiment regarding issues like wars [4], [5].

Furthermore, reports indicate that fake news regularly receives greater publicity on social media than actual news; current examples of such an occurrence may be found on well-known social media platforms. Compared to other forms of disinformation, false news is regarded as a more serious problem [6], [7]. Effective solutions to this problem are becoming

increasingly important as the spread of false news on social media raises concerns about the credibility of information accessible online. As the amount of data grows, it becomes increasingly important to capture meaningful data quickly and effectively. Because of the growing severity of this problem, researchers have been actively looking for ways to detect and prevent fake news. Because AI can systematically examine the smallest linguistic and contextual information that human moderators would overlook, it presents a particularly promising avenue for the employment of artificial intelligence (AI) in fake news [8], [9].

Recent breakthroughs in AI and natural language processing (NLP) have raised concerns in detecting fake news, which has prompted the establishment of various creative research approaches in this field of study [10]. Because machine learning (ML) and deep learning (DL) algorithms can recognize patterns and differentiate between authentic and fraudulent information using linguistic and statistical data, they have been widely used in the identification of false news. Building upon these advancements, this study proposes a model grounded in RoBERTa to specifically identify fake news in the textual content. Additionally, our work integrates explainable AI methods, ensuring that our processes are not only effective but also transparent and understandable,

aligning with the growing need for accountability in AI-driven solutions. The primary contributions of this work can be succinctly summarized as follows;

- In order to eliminate unnecessary elements from the textual data, different pre-processing operations are first carried out.
- For feature extraction, GloVe is used to generate meaningful word embeddings, capturing contextual relationships within the dataset.
- To optimize the extracted features, the Firefly Algorithm is employed for feature selection, ensuring that only the most relevant features contribute to classification.
- The extracted features are then fed into different hybrid deep learning models such as RoBERTa-BiLSTM, RoBERTa-GRU, and RoBERTa-LSTM for classification.
- To enhance overall performance, the hybrid deep learning models' predictions are integrated by majority voting.
- Besides, hyperparameters of LSTM and GRU are selected optimally using the Harris Hawks' optimization algorithm.
- Our model's classification is clarified using Explainable AI SHAP to further increase credibility and honesty.
- The model's performance is evaluated based on accuracy, precision, specificity, F1-score, and recall, exhibiting superiority over existing techniques.

The structure of the paper is organized as follows: Section 2 presents a review of the relevant literature, highlighting key studies and theoretical foundations. Section 3 discusses the research technique. Section 4 describes the results and discussion of the proposed model, while Section 5 concludes with a conclusion.

II. LITERATURE REVIEW

Hashmi et al. [11] demonstrated a trustworthy technique for identifying fake news by using three datasets, including FakeNewsNet, WELFake, as well as Fake News Prediction. Explainable AI employing LIME and LDA was used to support the transparency of the detection process and provide insightful insights. It was noted, although, that the weightiest term and several time-related terms may have deceived the model into believing the text to be an authentic news item. Using the RoBERTa transformer and LSTM architectures, Mohawesh et al. [12] introduced a unique semantic- and linguistic-aware model by integrating RoBERTa with an LSTM layer for fake review identification. Even though they achieved better accuracy, the model's reliance on computationally intensive architectures may hinder its real-time applicability in resource-constrained environments. Dhiman et al. [13] presented a model called Generative Bidirectional Encoder

Representations from Transformers (GBERT), focusing on the fake news classification issues by combining GPT and BERT. However, integrating two large-scale transformer models increases computing complexity, which limits its usefulness in resource-constrained environments. Jamjoom et al. [14] proposed robustly optimized bidirectional encoder representations based on the transformer technique for cyberbullying detection. GloVe features were combined with the RoBERTa model to improve cyberbullying prediction accuracy. However, the model suffers from high computational complexity, which may pose challenges in real-time deployment.

Abdelhakim et al [15] introduced an Arabic pre-trained BERT for the detection of Arabic fake news. They introduced a novel hybrid system that consists of two main parts. They implemented a hybrid architecture composed of two main components, such as Arabic Pre-trained BERT Transformer (APBT) and CNN. Even though they obtained greater performance, the absence of a preprocessing step to detect and rectify grammatical problems in input text could potentially degrade model performance further. Sastrawan et al. [16] developed a deep learning-based technique for detecting fake news, which included a data augmentation strategy to handle class imbalance. Ali et al. [17] created a deep ensemble model for detecting fake news by a sequential deep learning method. This suggested model was benchmarked using the two well-known and widely used datasets (ISOT and LIAR) with several classifiers. However, the method's reliance on handcrafted feature extraction techniques may limit its adaptability to diverse text structures. Umer et al. [18] introduced a method to enhance the predictive accuracy of cyberbullying detection in tweets. Their approach combined Principal Component Analysis (PCA) for feature extraction with GloVe word embeddings and the RoBERTa transformer model to capture both semantic and contextual information. Furthermore, to enhance model interpretability and provide insights into decision-making, LIME was incorporated. However, the integration of GloVe and RoBERTaNet may struggle to fully capture the nuanced, context-specific expressions often present in cyberbullying content, potentially limiting detection performance in subtle or ambiguous cases.

III. PROPOSED METHODOLOGY

3.1. Overview

The following subsection explains the proposed fake news detection method, implementing an ensemble hybrid deep learning model with the LIAR dataset, which includes properties such as statements, dates, subjects, and other contextual variables. Initially, several pre-processing processes are conducted to remove redundant elements from the textual material. For feature extraction, GloVe is used to

generate meaningful word embeddings, and to optimize the extracted features, the Firefly Algorithm is employed for feature selection. For feature extraction and classification, the retrieved features are fed into three models, such as RoBERTa-BiLSTM, RoBERTa-LSTM, and RoBERTa-GRU. To enhance overall performance, the hybrid deep learning models'

predictions are aggregated by majority voting. Additionally, the HHO technique is employed to choose the LSTM and GRU hyperparameters in the best possible way. We use Explainable AI SHAP to elucidate our model's classification to increase transparency and believability. The general structure of the suggested methodology is given in Figure 1.

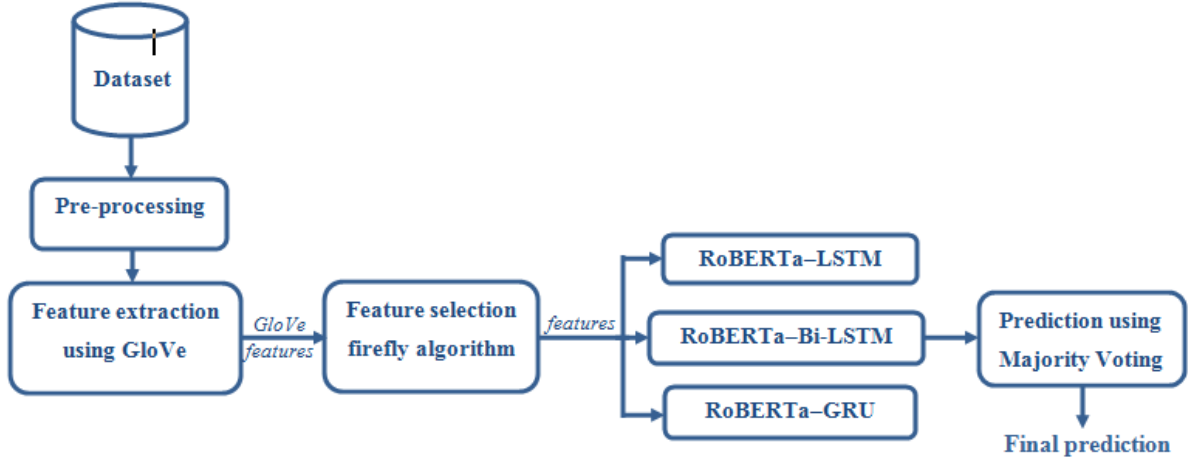


Figure 1: General Structure of the Proposed Methodology

3.2. Dataset Pre-processing

The LIAR dataset, which consists of political statements and associated metadata, often contains inconsistencies, redundancies, and irrelevant textual elements. Hence, the collected raw data must be cleaned before being translated into a format that can be processed further. Here, three methods, such as stop word removal, stemming, and lemmatization, are used in this study to refine the data in the LIAR dataset.

3.3. Feature Extraction and Selection

To extract contextual features from the text of statements in the LIAR dataset, GloVe is employed in this study. An unsupervised learning technique called GloVe compares two words' distances in a vector space to how close they are to one another. Word embeddings are the vector representations that are produced. Embedding Words GloVe builds a corpus that will illustrate the connections between words and generates a co-occurrence matrix. The formulations for GloVe are as follows:

$$G_i^T + \vec{G}_k + b_i + \vec{b}_k = \log(Z_{ik}) \quad (1)$$

Where, G_i is a word vector, $\vec{G}_k$ is a word context vector, b_i represents major scalar bias, $\vec{b}_k$ represents word context scalar bias and Z denotes emergence matrix.

The semantic features extracted using GloVe embeddings are further refined through the application of the FA for optimal feature selection,

which enhances the effectiveness of fake news detection. Inspired by the flashing activity of fireflies in their natural habitat, Xin-She Yang [19] introduced the FA. The algorithm's performance is based on actual firefly behavior, which depends on the attraction between two fireflies, dependent on brightness. The light intensity and the fireflies' attraction to one another are two crucial components of the firefly algorithm. The brightness of the firefly, which is expressed and computed using a sort of fitness function, determines how much light is present in each source. The approach represents every possible solution to the optimization problem as a firefly in the search zone, with every firefly's brightness indicating the caliber of its solution. The fireflies are then iteratively moved towards brighter ones by the algorithm, which takes into account their relative distances and mutual attraction. The system effectively searches the area for the best results by adjusting the firefly's positions based on these instructions. The FA iterates through position updates and an attraction mechanism until a predefined stopping criterion is met. At termination, FA returns the optimal feature subset that maximizes detection performance while minimizing redundant features.

3.4. Classification using an Ensemble Hybrid Deep Learning Model

The features, derived from GloVe embeddings and refined through the FA, are input into the proposed ensemble model. An ensemble hybrid deep learning model is employed to classify fake news using the selected semantic features. The suggested ensemble model is made up of three hybrid deep learning models. These models combine the RoBERTa and

sequence models, which include LSTM, Bi-LSTM, as well as GRU. The three hybrid variants are RoBERTa-BiLSTM, RoBERTa-LSTM, and RoBERTa-GRU.

3.4.1 Robustly Optimized BERT Approach

RoBERTa is a refined transformer-based model that enhances the original BERT architecture by refining its pre-training methodology. RoBERTa appears as a valuable tool in the field of NLP, designed to improve the efficacy of natural language understanding (NLU) tasks. RoBERTa, with its 12-layer design and 768 hidden states per layer, tries to overcome the shortcomings of its predecessor by combining intensive pretraining and fine-tuning procedures. RoBERTa trains on ten times larger datasets over a greater amount of time, using larger batches as well as longer sequences than BERT. RoBERTa was trained on 4 datasets: the English Wikipedia and Book Corpus datasets, the CommonCrawl (CC)-News dataset, the OpenWebText dataset, and the Stories dataset.

In the suggested hybrid model, the pretrained RoBERTa model serves as the encoder, tokenizing the input text, effectively converting tokens into meaningful word embedding forms. The pretrained RoBERTa model creates these word embeddings, which are subsequently entered into a sequence model to identify long-range dependencies within the word embedding sequence.

3.4.1.1 RoBERTa-LSTM

The word embeddings, generated by the pretrained RoBERTa model, are then fed into an LSTM layer to capture long-range dependencies within the sequence of word embeddings. A sequence model called LSTM was developed to address the vanishing gradient and long-short-term memory issues with conventional RNNs. To enable LSTM to encode the input's long-range dependencies, three gates are crucial to its gating process. The forget, input, and output gates are the three types of gates. To start, the forget gate is utilized to decide which appropriate information needs to be discarded or retained. Second, the input gate is used to determine which value at present should be updated. The output gate will ultimately determine which data should be sent to the following hidden state. The computations performed in the LSTM unit are shown below:

$$F_t = \sigma(\omega_F z_t + u_F H_{t-1} + b_F) \quad (2)$$

$$I_t = \sigma(\omega_I z_t + u_I H_{t-1} + b_I) \quad (3)$$

$$O_t = \sigma(\omega_O z_t + u_O H_{t-1} + b_O) \quad (4)$$

$$\tilde{C}_t = \tanh(\omega_C z_t + u_C H_{t-1} + b_C) \quad (5)$$

$$C_t = C_{t-1} \times F_t + I_t \times \tilde{C}_t \quad (6)$$

$$H_t = O_t \times \tanh(C_t) \quad (7)$$

where the input gate, forget gate, output gate, hidden state, and cell state at the current time step t given the input z are indicated by I_t, F_t, O_t, H_t and C_t , respectively. Three of the gates contain the sigmoid function applied to them, which helps determine if they permit or block input at this particular time step. If the output value of the sigmoid function is close to one, information is capable of being communicated to the next state; if it is close to zero, information cannot be transmitted. Furthermore, the weight matrices and biases in the input gate, forget gate, cell state, and output gate are represented by $(\omega_I, \omega_F, \omega_C, \omega_O, u_F, u_I, u_C, u_O)$ and (b_I, b_F, b_C, b_O) respectively. Because the hybrid RoBERTa-LSTM model contains 256 LSTM units, the computations will be repeated 256 times during each training session.

3.4.1.2 RoBERTa-BiLSTM

The word embeddings are generated by the pretrained RoBERTa model and then put into a Bi-LSTM layer to find long-range relationships in the embedding sequence. Every time step, the bidirectional LSTM performs forward as well as backward passes. This allows the Bi-LSTM to gather information from both the past and the future. The hybrid RoBERTa-BiLSTM model uses an LSTM unit of 128 in both directions.

3.4.1.3 RoBERTa-GRU

The pretrained RoBERTa model creates the word embeddings, which are subsequently input into a GRU layer to identify long-range relationships in the word embedding sequence. The GRU is another sequence model designed to address RNN vanishing gradient concerns. GRU's more simplistic architecture also employs gating mechanisms to describe long-range interactions. Additionally, there are two gates in GRU: the reset gate and the update gate. The update gate determines whether information ought to be communicated forward in the future, whereas the reset gate erases unneeded information from the input. The GRU unit's calculations are described as follows:

$$K_t = \sigma(\omega_K z_t + u_K H_{t-1} + b_K) \quad (8)$$

$$R_t = \sigma(\omega_R z_t + u_R H_{t-1} + b_R) \quad (9)$$

$$C_t = \tanh(\omega_C z_t + u_C (R_t * H_{t-1}) + b_C) \quad (10)$$

$$H_t = K_t * H_{t-1} + (1 - K_t) * C_t \quad (11)$$

where R_t and K_t denote the reset gate and update gate.

3.4.2 Hyperparameter Optimization using Harris Hawks Optimization

To enhance the performance of the classifier, we are employing HHO to tune the hyperparameters of LSTM and GRU. In 2019, Heidari et al. introduced a novel meta-heuristic optimization technique called HHO. It draws inspiration from Harris's hawk's distinctive cooperative foraging habits. Depending on how the environment changes and how the prey escapes, Harris's hawks are capable of participating in a variety of pursuit actions. These coordinated tactics can aid Harris's hawk chase the discovered prey until it is exhausted, increasing its vulnerability. These switching behaviors also help to confuse the running prey. This intelligent switching enhances the optimizer's ability to avoid local optima and find better global solutions for hyperparameter settings.

The procedures for optimizing the hyperparameters in LSTM and GRU are as follows:

3.4.2.1 Initialization

Harris's hawks are responsible for initializing the population, commonly known as the Q matrix. Consequently, each Harris's hawk represents a vector-based solution. In this method, the hyperparameters in the LSTM, such as Batch Size (β), Learning rate (α), Epochs (γ), Dropout Rate (ϕ) and the hyperparameters in the GRU, such as batch size (ϕ), learning rate (η), no. of epochs (ψ), are regarded as potential solutions and are initialized as follows: Here is the format of the initial solution;

$$Y_i = \{y_1, y_2, \dots, y_n\} \quad (12)$$

Here, Y_i indicates the population of the solution, and n denotes the parameters,

$$y_1 = \{\alpha_1, \beta_1, \gamma_1, \phi_1\} // \text{hyper-parameters of LSTM} \quad (13)$$

$$y_1 = \{\eta_1, \phi_1, \psi_1\} // \text{hyper parameters of GRU} \quad (14)$$

3.4.2.2 Fitness Function

The fitness function evaluates each candidate solution by considering both the initialized candidate solution and its opposite solution. At each iteration, the fitness value for $f(y_i)$ is computed to determine the optimal selection. The fitness function is mathematically formulated as below:

$$\text{fitness} = \text{Max}(\text{accuracy}) \quad (15)$$

3.4.2.3 Exploration Phase

Typically, Harris's hawks wait and scan the desert for prey while perching in different positions. Two perching techniques are determined by the random tall trees or the positions between additional family members and the prey, which are chosen based on the random q value.

$$y(t+1) = \begin{cases} y_r(t) - r_1 |y_r(t) - 2r_2 y(t)| & q \geq 0.5 \\ (y_{prey}(t) - y_m(t)) - r_3(lb + r_4(ub - lb)) & q < 0.5 \end{cases} \quad (16)$$

$$y_m(t) = \frac{1}{n} \sum_{i=1}^n y_i(t) \quad (17)$$

where $y_r(t)$ represents the random hawk's location, $y_{prey}(t)$ is the prey's location, which is the optimal position recorded thus far, then $y_m(t)$ is means location at the moment. The amount of hawks is represented by n , the upper and lower bounds of the issue are ub and lb , and the random values and q , r_1 , r_2 , r_3 , and r_4 range from 0 to 1.

3.4.2.4 Transition from Exploration to Exploitation Phase

The HHO technique uses a transition strategy between exploration and exploitation based upon the prey's escaping energy, which then adjusts the numerous exploitative activities. The energy of the prey can be observed in the following manner, and it decreases throughout the escape behavior.

$$E_e = 2E_{e0} \left(1 - \frac{t}{T}\right) \quad (18)$$

where E_e stands for the prey's escape energy, E_{e0} for the beginning state of the energy, T is the maximum number of iterations. The optimization algorithm completes the exploration stage whenever $|E_e| \geq 1$ occurs, and the exploitation phase when $|E_e| < 1$ occurs.

3.4.2.5 Exploitation Phase

Based on the prey's energy for escape as well as the Harris's hawks' defense techniques, four distinct attacking and chasing tactics are suggested in this phase. In addition to the fleeing energy, the chasing strategy is chosen using parameter r , which shows the likelihood that the prey will escape ($r < 0.5$) or not ($r \geq 0.5$) before attack.

Soft besiege

When $r \geq 0.5$ and $|E_e| \geq 0.5$, the Harris's hawks gently encircle it to wear it down before attacking. The following is an example of this behavior:

$$y(t+1) = \Delta y(t) - E_0 |Jy_{prey}(t) - y(t)| \quad (19)$$

$$\Delta y(t) = y_{prey}(t) - y(t) \quad (20)$$

$$j = 2(1 - r_5) \quad (21)$$

where $\Delta y(t)$ is the difference between the prey's previous and current positions, j represents the prey's random jump resilience, $y_{prey}(t)$ is the prey's location, $y(t)$ represents the prey's current location, as well as r_5 denotes a random number that ranges from 0 to 1.

Hard besiege

The prey had minimal escaping energy when $r \geq 0.5$ and $|E_e| < 0.5$, which makes it easy for the Harris's hawks to eventually attack it. The following is an update to the locations in this scenario:

$$y(t+1) = y_{prey}(t) - E|\Delta y(t)| \quad (22)$$

Soft besiege with progressive, rapid dives

Since the prey possesses the proper energy to escape when $|E_e| \geq 0.5$ and $r < 0.5$, Harris's hawks engage in a soft siege, making multiple quick dives around the prey while attempting to gradually adjust their location and direction. The following is an example of this behavior:

$$y_{t+1} = \begin{cases} X : y_{prey,t} - E_e |Jy_{prey,t} - y_t|, & \text{if } f(X) < f(y_t) \\ Z : X + s \times LF(d), & \text{if } f(Z) < f(y_t) \end{cases} \quad (23)$$

where s represents a random vector and d indicates the problem's dimension size. The Levy flight function, or LF, is used to imitate the prey's deceptive movements. Take in consideration that the only position chosen for the following position is the better one among X and Z . Hard besiege with progressive rapid dives

The prey lacks the energy to flee when $|E_e| < 0.5$ and $r < 0.5$, therefore the hawks engage in a rigorous besiege to close the gap between their average location and the prey before attacking and killing it. The mathematical depiction is given as follows:

$$y_{t+1} = \begin{cases} X : y_{prey,t} - E_e |Jy_{prey,t} - y_{m,t}|, & \text{if } f(X) < f(y_t) \\ Z : X + s \times LF(d), & \text{if } f(Z) < f(y_t) \end{cases} \quad (24)$$

Termination:

Each iteration of the suggested HHO algorithm updates the selection of optimal hyperparameters by the three phases. The final output of the HHO algorithm is the optimal hyperparameters that maximize the model's performance while minimizing overhead.

3.4 Prediction using Majority Voting

The proposed strategy uses three pre-trained classification models, namely RoBERTa-BiLSTM, RoBERTa-LSTM, and RoBERTa-GRU. Each of these models is designed to learn complex relationships between the contextual and temporal features extracted from the LIAR dataset and to independently classify each statement as either FAKE or TRUE. This method uses a majority-based voting technique to improve classification results. Each of the chosen models calculates the classification results for each test case separately, and its final result is forecasted using the results that achieve majority representation. In the case of majority voting, each of the classifiers' majority (plurality) consensus, indicated by the C , predicts the class label y .

$$\hat{y}_{final} = Mode\{C_1(y), C_2(y), C_3(y)\} \quad (25)$$

Where $Mode$, returns the class label that appears most frequently among the predictions.

IV. RESULTS AND DISCUSSIONS

This section gives an extensive evaluation of the experiments carried out, comprising the description of the datasets, metrics for evaluation, and the efficacy of the proposed approach. It also provides a performance evaluation of the suggested model with alternative models. Our model's classification is clarified using Explainable AI SHAP to further increase transparency and believability. Using the Python language, the system was implemented. The implementation requires a CPU-based computer system with 8 GB of RAM, 256 GB of memory and a 2 GHz Intel Core i7 processor.

4.1 Dataset description

The proposed models were trained and tested on a benchmark dataset termed as "LIAR Dataset". LIAR is a publicly accessible dataset for detecting fake news. Politifact.com collected 12.8K hand-labeled brief statements in diverse contexts over a decade, and each case includes a thorough analysis report and links to source documents. Politifact.com's news assertions were compiled from a variety of sources, including television advertising, Facebook posts, Twitter, interviews, political discussions, and so on. Politifact.com classifies the statements in the dataset into six categories: false, pants-fire false, true, half-true, largely true, and barely true.

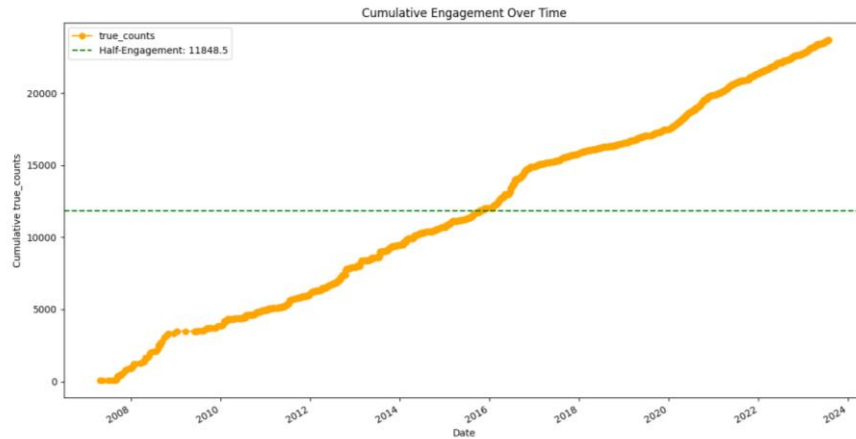


Figure 2: Cumulative Engagement over Time-Test

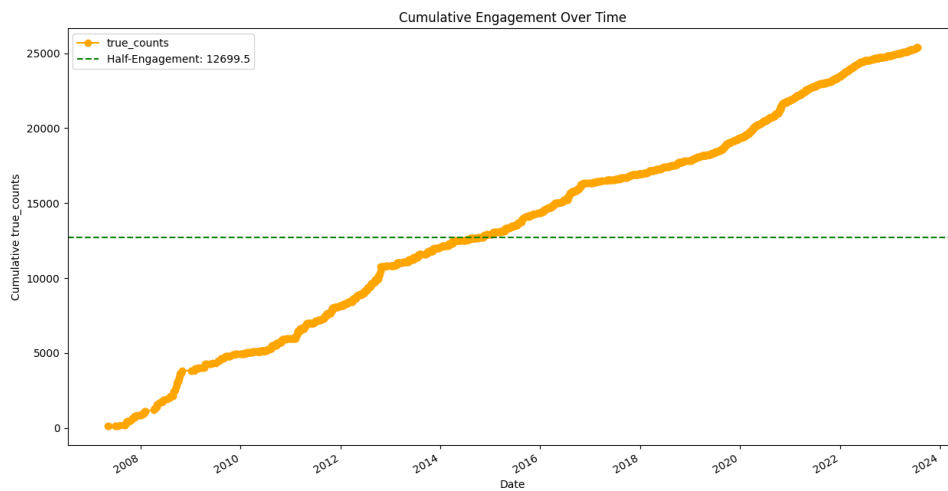


Figure 3: Cumulative Engagement over Time-Train

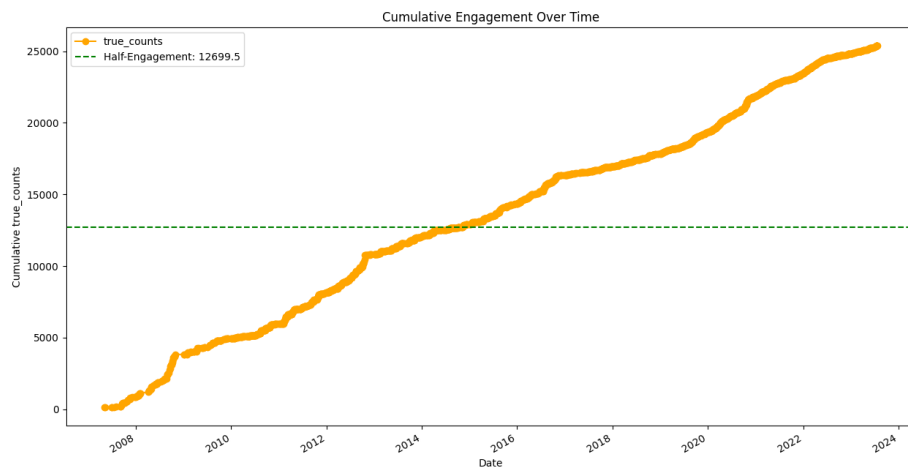


Figure 4: Cumulative Engagement over Time- Valid

Figures 2, 3 and 4 represent the cumulative engagement over time (test, train, and validation) based on the LIAR dataset. The y-axis represents the cumulative number of true instances (true_counts), which could correspond to factually correct statements as labeled in the dataset. The x-axis shows the timeline from 2008 to 2024. The steadily increasing orange curve indicates how the number of truth-labeled data points has grown over time. The

green dashed line marks the halfway point of cumulative engagement (11,848.5 for test and 12,699.5 for train and valid), showing that more than half of the true-labeled content was recorded before 2016. This trend provides context for the LIAR dataset, revealing how engagement with factual statements has evolved, and it may help in understanding temporal biases or data distribution shifts in fake news detection models.

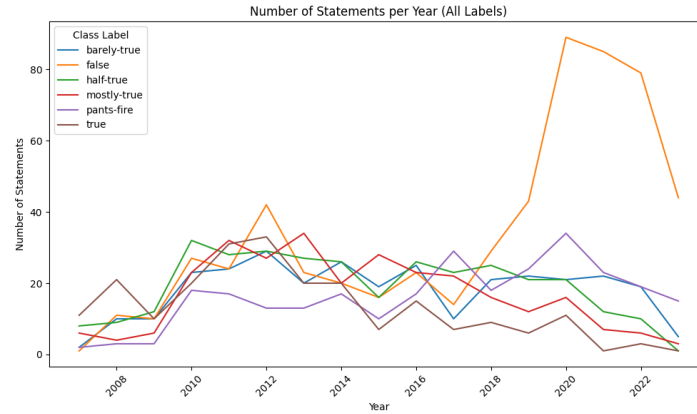


Figure 5: Distribution of statements across different class labels (training)

Figures 5, 6 and 7 illustrate the year-wise distribution of statements across six different class labels: true, false, mostly-true, barely-true, half-true, and pants-fire. From 2008 to 2022, the number of statements in each class fluctuates with noticeable trends. Each line represents how the number of statements varies particular time. If the slope increases, it indicates that

the specific class is growing in the particular period. If it is decreasing, it indicates that the specific class is decreasing at that time. This temporal variation provides useful information on how the distribution of truthful and fake statements changes over time.

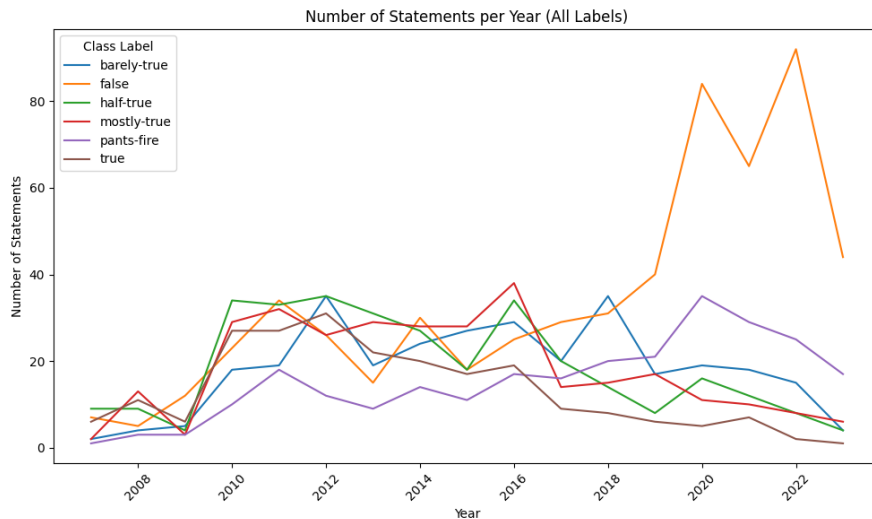


Figure 6: Distribution of statements across different class labels (testing)

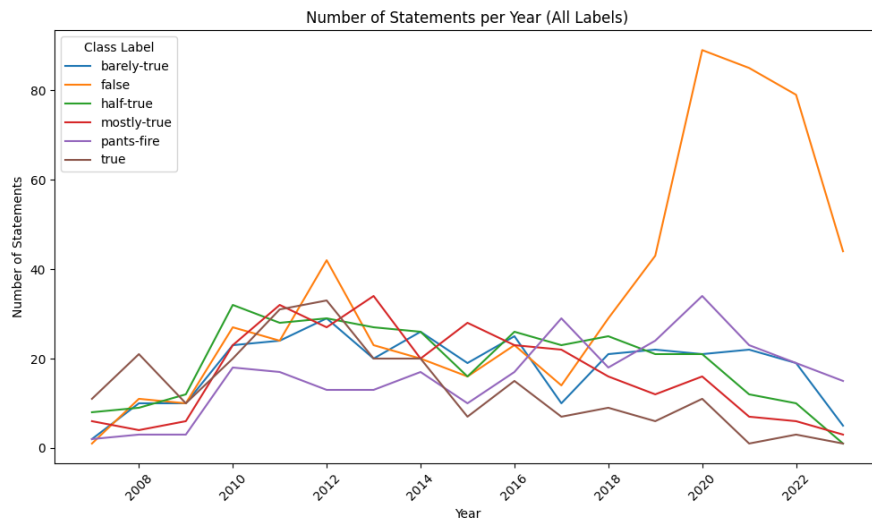


Figure 7: Distribution of statements across different class labels (validation)

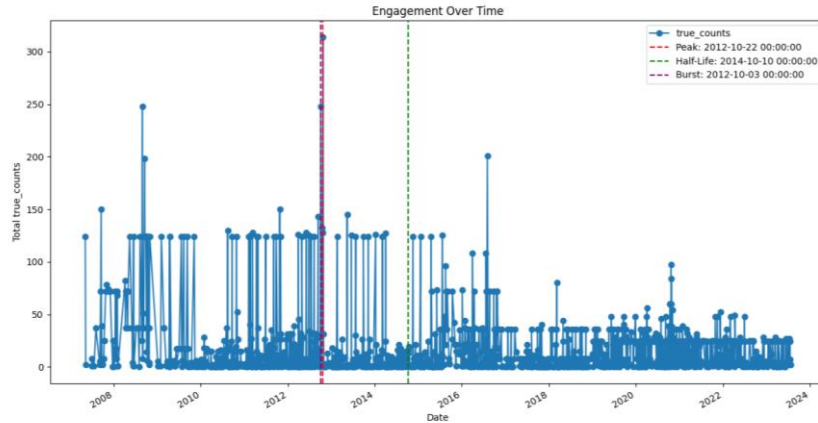


Figure 8: Engagement over Time (training)

Figures 8, 9, and 10 show the Engagement over Time, indicating user interactions like views, likes, and other types of engagement for the test, train, and validation datasets. Each data point reflects the limit of engagement at a specific time, provides information about how interactions are obtained. The red vertical line reflects the highest engagement date (200809–26), which symbolizes the highest interaction time. The green line is the half-life point (2015–10-15), indicating that when the conversation fell to 50% of its total, the natural fading of the conversation is reflected. The purple line (2012-10-

16) indicates a sudden spike in the conversation, which may be caused by a particular event or amplification on social media, indicating a burst of interest. Inspection of these temporary trends provides insight into the lifetime and virality of information, and it is particularly helpful in research on the spread of fake news. Identifying when and how engagement spikes or diminishes can support more accurate detection and modeling of misleading content propagation patterns.

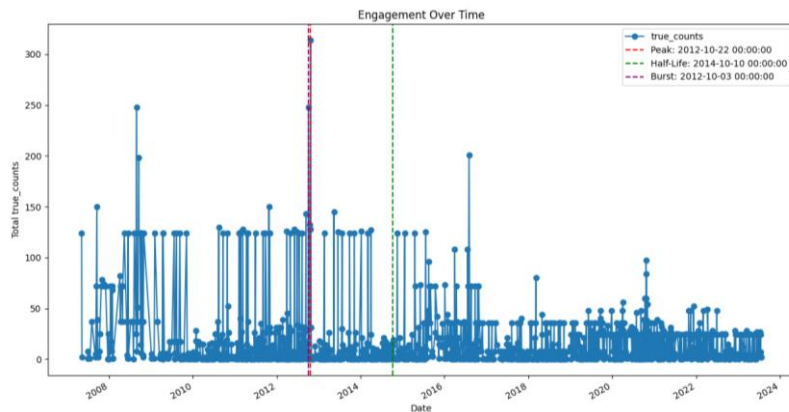


Figure 9: Engagement over Time (validation)

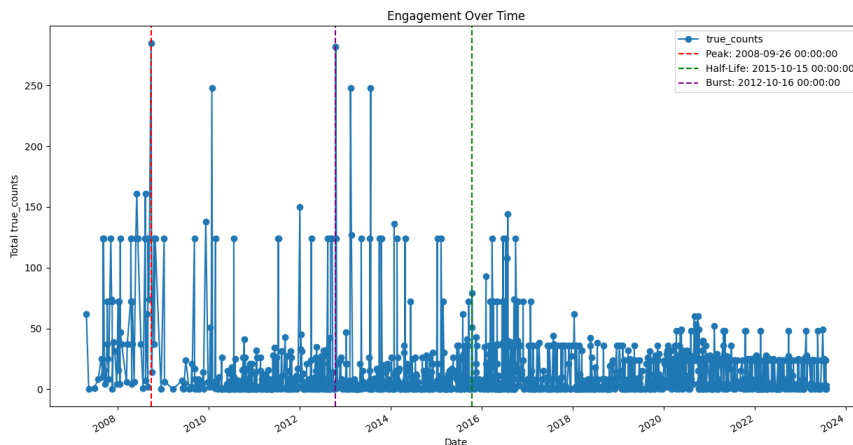


Figure 10: Engagement over Time (testing)

4.2 Performance Evaluation of the proposed ensemble approach with other methods

This section compares the effectiveness of our proposed RoBERTa with sequence models with other methods, such as BERT with sequence models and

XLNet with sequence models, using the LIAR dataset in terms of Accuracy, Precision, F-Measure, Recall, and Specificity. Table 2 depicts the comparative analysis of proposed and existing methods.

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Specificity (%)
Proposed RoBERTa with a sequence model	98.9	98.58	99.18	98.88	98.62
BERT with a sequence model	96.4	96.24	96.44	96.34	96.35
XLNet with a sequence model	93.55	94.02	92.78	93.40	94.29

Table 1: Comparative Analysis of Proposed and Existing Methods

Table 2 shows the results of fake news detection through the Proposed RoBERTa with sequence models, with other existing models, including accuracy, F-Score, recall, precision, and specificity. Our proposed method outperforms the BERT and XLNet models in terms of all the evaluation metrics. Our experimental results demonstrate that the proposed RoBERTa with sequence models outperforms BERT by 2.5% in accuracy, 2.54% in F-score, 2.34% in precision, 2.74% in recall, and 2.27% in specificity. Compared to the XLNet model, the proposed model achieves notable improvements of 5.35% in accuracy, 5.48% in f-score, 4.56% in precision, 6.4% in recall, and 4.33% in specificity, respectively. These outcomes highlight the efficiency of the proposed method in achieving a superior balance of accuracy, recall, precision, and specificity, making it more reliable for fake news detection than the other compared models.

4.3 Interpretability Analysis Using SHAP for Fake News Detection

Since interpretability modeling facilitates the creation of models or tactics that make intricate machine learning procedures more transparent and intelligible,

it is essential within the area of fake news identification. In this work, we use SHAP to clarify our model's classification. SHAP is one of the additive feature attribution value methods initially introduced by Lundberg and Lee (2017). It was created for explainable AI. The explanation level focuses on understanding how a model makes decisions based on its features and the learned components. SHAP is a class of model-agnostic additive feature attribution algorithms that assign value to each input feature in any machine learning or deep learning model. The SHAP explanation for a model prediction classified as fake news is displayed in Figure 8. Every token in the input sentence has a feature contribution value, which represents its effect on the model's prediction. Positive impact values indicate a bias towards the "fake news" class, and negative values represent a bias towards "real news." As an illustration, the token "Michelle" had a high positive effect value and most likely contributed significantly to classification. The general model output value was 0.344, and since the aggregate of the contributions made it higher than the cut-off value, the prediction went to fake news.

```

Feature Contributions:
[CLS]: 4.0978193283081055e-08 → Positive impact (slightly towards fake news).
michelle: 0.11463629268109798 → Strong positive impact (likely fake news).
nun: 0.055331128649413586 → Positive impact (slightly towards fake news).
#n: 0.04949651565402746 → Positive impact (slightly towards fake news).
has: -0.004016709979623556 → Negative impact (slightly towards real news).
praised: 0.016335456166416407 → Positive impact (slightly towards fake news).
the: 0.009028233354911208 → Positive impact (slightly towards fake news).
occupy: 0.018055879278108478 → Positive impact (slightly towards fake news).
movement: 0.004004559013992548 → Positive impact (slightly towards fake news).
.: 0.08149681612849236 → Positive impact (slightly towards fake news).
[SEP]: 5.21540641784668e-08 → Positive impact (slightly towards fake news).

Base Value: Could not be calculated.
Model Output Value: 0.34436826407909393

Explanation of Prediction:
The model predicted 1 (fake news). This is because the sum of the feature contributions, when added to the base value, resulted in a value above the decision threshold.

```

Figure 8: SHAP Explanation Output 1

Figure 9 shows the SHAP explanation of another prediction labeled as fake news. The feature contribution for each token is its impact on the prediction. Tokens like "trump", "health", and "medicare" had different impacts. Whereas there

were negatively contributing tokens such as "Medicare" and "pick" that somewhat supported a classification of real news, there were positively contributing tokens such as "Trump" and "wants" that helped drive the prediction towards fake news. The

value of the model output was 0.058, and while this is quite low, the overall contribution somehow kept the

decision crossing the classification threshold for fake news.

```
Feature Contributions:
[CLS]: 3.166496753692627e-08 → Positive impact (slightly towards fake news).
trump: 0.005709423683583736 → Positive impact (slightly towards fake news).
': -0.004187919199466705 → Negative impact (slightly towards real news).
s: 0.011789320968091488 → Positive impact (slightly towards fake news).
health: 0.0035262289457023144 → Positive impact (slightly towards fake news).
secretary: 0.007278262171894312 → Positive impact (slightly towards fake news).
pick: -0.007002951577305794 → Negative impact (slightly towards real news).
wants: 0.01768120378255844 → Positive impact (slightly towards fake news).
a: 0.011447340250015259 → Positive impact (slightly towards fake news).
medicare: -0.019550835713744164 → Negative impact (slightly towards real news).
phase: 0.004381263628602028 → Positive impact (slightly towards fake news).
#out: 0.001571696251630783 → Positive impact (slightly towards fake news).
by: -0.00428311713039875 → Negative impact (slightly towards real news).
summer: 0.00920926500118183 → Positive impact (slightly towards fake news).
2017: 0.01436705980449915 → Positive impact (slightly towards fake news).
.: 0.006223838776350021 → Positive impact (slightly towards fake news).
[SEP]: -4.470348358154297e-08 → Negative impact (slightly towards real news).

Base Value: Could not be calculated.
Model Output Value: 0.0581606660461426

Explanation of Prediction:
The model predicted 1 (fake news). This is because the sum of the feature contributions, when added to the base value, resulted in a value above the decision threshold.
```

Figure 9: SHAP Explanation Output 2

V. CONCLUSION

This study proposed a robust ensemble deep learning framework for fake review detection, which integrates a RoBERTa model with sequential models such as LSTM, Bi-LSTM, and GRU. The model employed GloVe word embeddings to capture rich contextual and semantic relationships in the textual data. To optimize the extracted features, the Firefly Algorithm was employed for feature selection, ensuring that only the relevant features contribute to classification. The ensemble classification was done by majority voting among RoBERTa-LSTM, RoBERTa-BiLSTM, and RoBERTa-GRU models. In addition, hyperparameters of LSTM and GRU were optimally chosen using Harris Hawks' optimization algorithm. To make it more transparent and credible, Explainable AI SHAP was used to explain the classification of our proposed model. The performance of our suggested model was measured based on accuracy, specificity, F1-score, precision, and recall, with superiority over other techniques like BERT and XLNet. Our experimental outcomes revealed that the proposed RoBERTa with sequence models performs better than the BERT 2.5% in accuracy and the XLNet model by 5.35% in accuracy. The model was tested with a single dataset that could restrict the applicability of the findings across various domains or types of review data. The future work should validate the model on more than one dataset from diverse domains to examine its resilience.

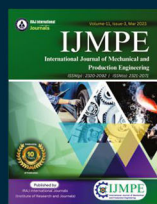
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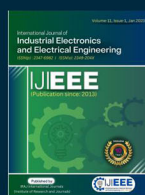


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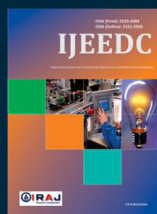


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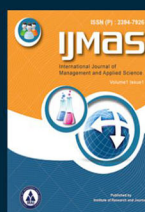


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